

Computational Neuropathology: Clinical and Research Applications of Computer Vision Models

Phedias Diamandis, MD, PhD, FRCPC

Neuropathologist, University Health Network

Scientist, Princess Margaret Cancer Centre

Associate Professor

Department of Laboratory Medicine & Pathobiology

University of Toronto

p.diamandis@mail.utoronto.ca

Lab: www.diamandis.org



UHN

Laboratory
Medicine
Program

Princess
Margaret
Cancer Centre

Disclosures

- I have no relevant financial relationships to disclose

Learning Objectives

1. Discuss concepts of how AI can be used to automate classification and quantitative tasks in neuropathology
2. Discuss how quantitative tissue feature signatures can be applied to predict outcomes in molecular subtypes of brain tumors
3. Describe how quantitative histology can aid in the study of intra-tumoral heterogeneity
4. List limitations and caveats of computational techniques in histological analysis

Outline & Disclaimer

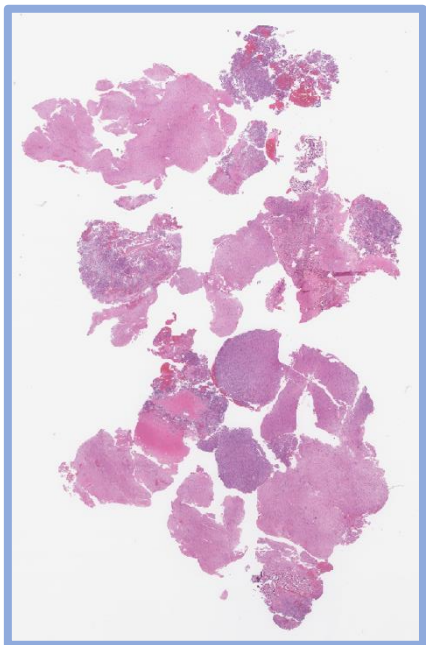
- Semantic Segmentation:
 - Supervised Learning
 - H&E-based tumor detection and classification
- Instance Segmentation:
 - IHC-based : Ki-67 Quantification
- Multi-slide Image registration
 - Merging H&E and IHC-level information
- Additional applications:
 - Unsupervised Learning: Heterogeneity assessment
 - Custom model development & clinical correlation studies
 - PHARAOH (pathologyreports.ai)
 - Integrating and refining molecular subtypes of gliomas

Note of Caution

- Not exhaustive survey of field
- Engineering NOT a knowledge-based field
- Focus on concepts and tools used / developed first-hand and not necessarily on "Best" point solutions for each problem
- Results vary based on study design

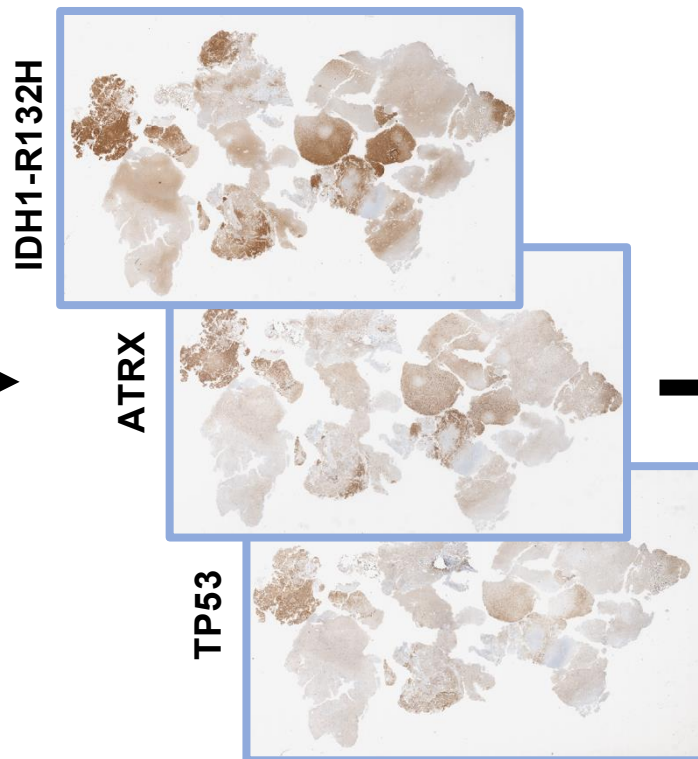
Modern molecular neuropathology workflow

Step 1 Histomorphologic Classification



H&E slide

Step 2 Biomarker Interpretation and Integration



Step 3 Integrated Diagnosis

University Health Network LABORATORY MEDICINE PROGRAM			
DEPARTMENT OF PATHOLOGY 200 Elizabeth Street Toronto, Ontario, M5G 2C4 TEL: 416-340-3325 FAX: 416-586-9901			
Surgical Pathology Consultation Report * Added *			
Patient Name:	Patient, USCAP	Service:	TGH Thoracic
MRN:	0876543	Visit #:	23412312345
DOB:	11/22/1947 (Age: 68)	Location:	2C Pre Operative Care Unit
Gender:	F	Facility:	TGH PMH
Referring MD:	1234567890		
Ordering MD:	Deep C. Friend, MD		
Copy To:	Good P. Friend, MD		
Specimen(s) Received:	1. Lymph-Node: ST10R TB Angle 2. Right middle lobe 3. Station 11R 4. Station 6R 5. Station 7 6. Interlobar ST11 7. Right middle and upper lobectomy		
Consolidated Therapeutic Report			
Interpretation			
Invasive moderately differentiated adenocarcinoma, acinar-predominant, pT2aN1 - POSITIVE for EGFR L858R mutation (see Molecular Diagnostics report) - NEGATIVE for ALK by immunohistochemistry (performed using the 5A4 antibody with a protocol optimized for detection of ALK gene rearrangement) - See Diagnosis, Comment, and Synoptic Report below for further details			
Signed out by: Lung Path, MD Date Reported: Jun-01-2018			
Diagnosis			
1.3-4. Lymph nodes (ST10R right tracheobronchial, ST11R right interlobar, ST4R right lower paratracheal, ST7 subcarinal, ST11 interlobar): - At least one lymph node per station, negative for malignancy (x5) (0/5) 2. Lung, resection (right middle lobectomy): a. Invasive moderately differentiated adenocarcinoma, acinar-predominant, pT2aN1, with: i. Greatest tumor dimension: 1.2 cm (see Comment) ii. Visceral pleural and lympho-vascular invasion present b. Staged parenchymal resection margin positive for carcinoma (see Comment) c. One of five lymph nodes locally positive for adenocarcinoma by direct invasion (1/5) (see Comment)			
Patient, USCAP Page 1 of 5			

Advantages:

- Excellent turnaround
- Low tissue requirements
- Cost effective

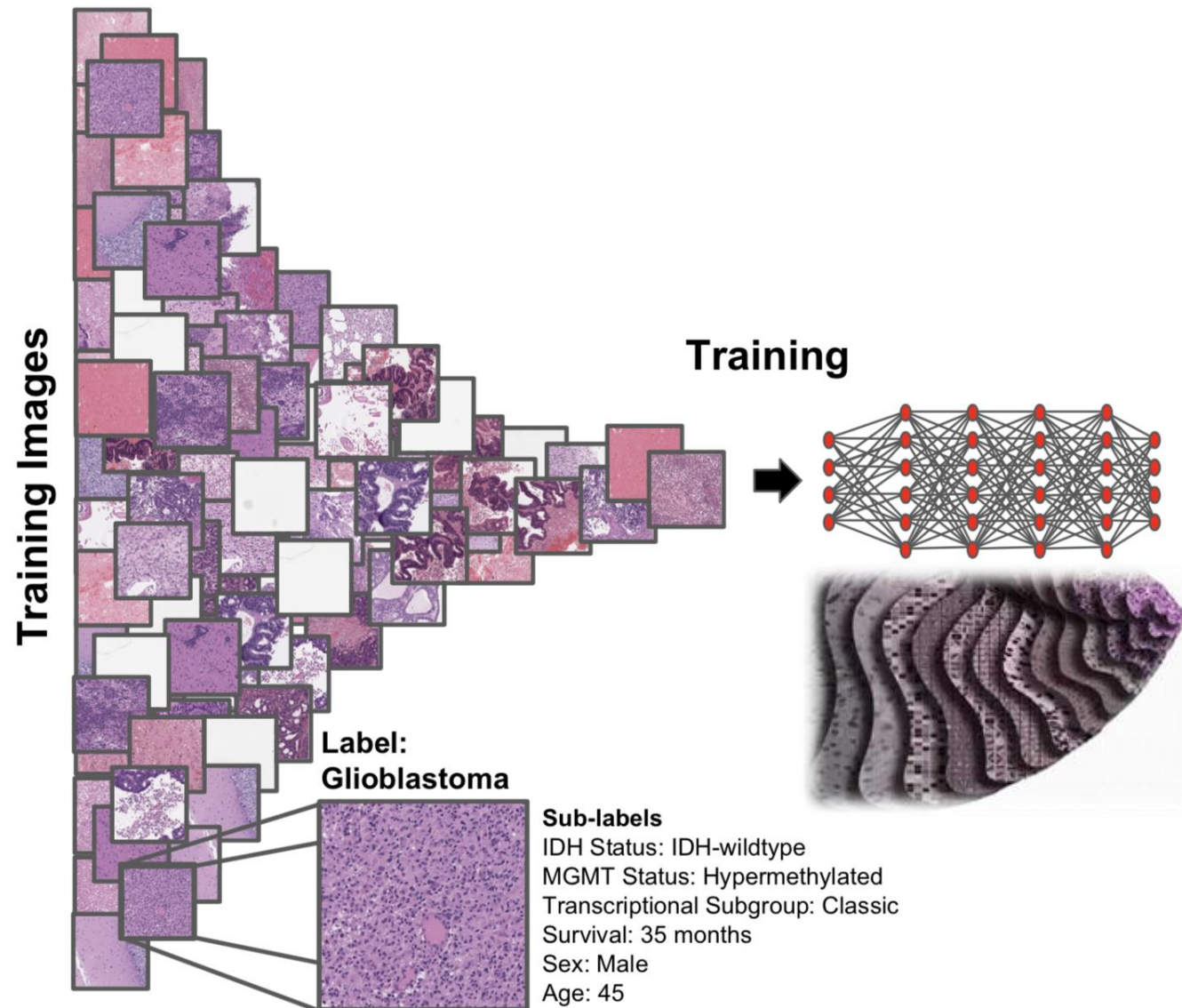
Disadvantages:

- Many manual steps
- Qualitative

Advanced
molecular studies

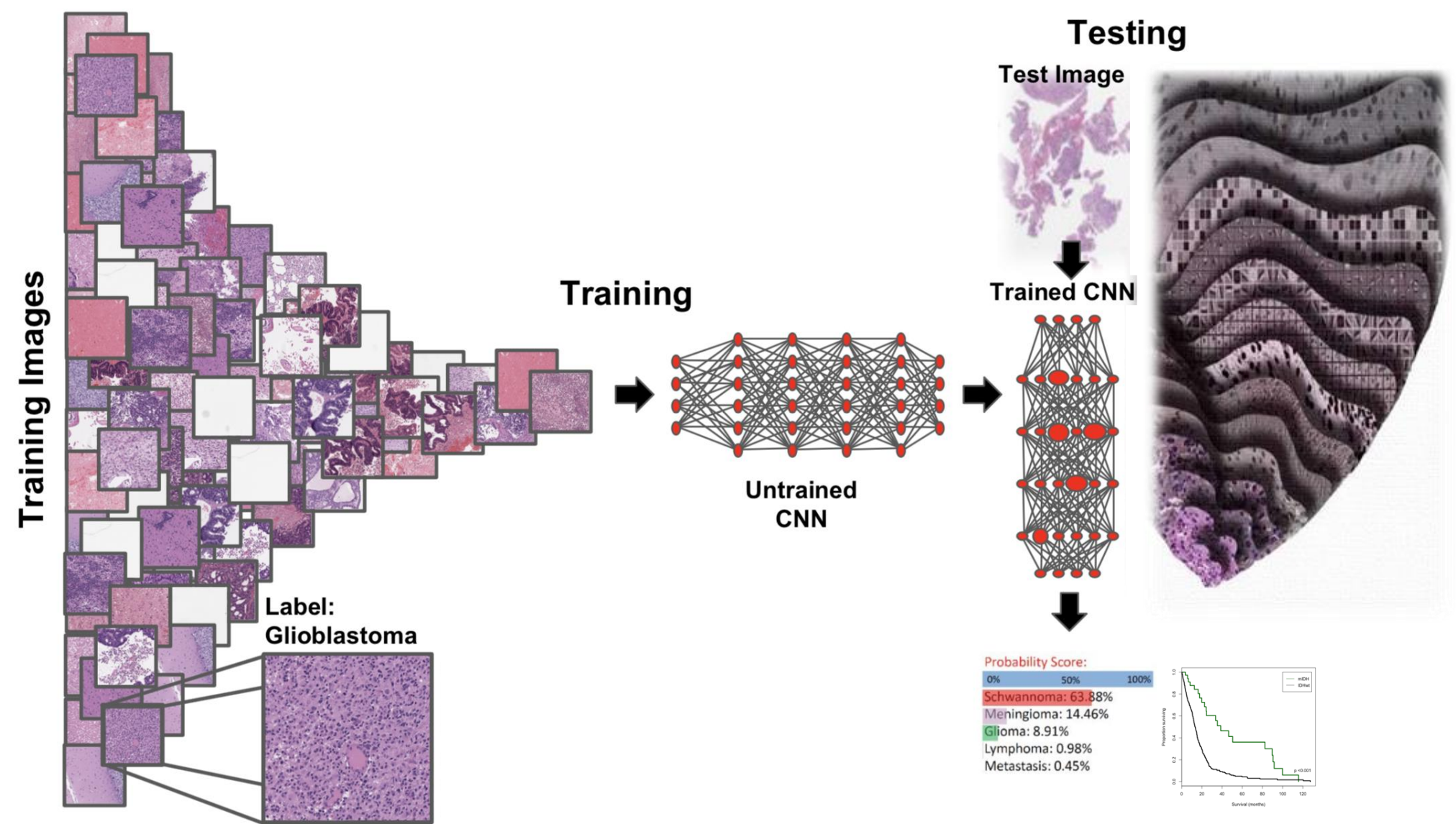
Step 1: Semantic segmentation

Neural Networks - Computer vision tools for automated pattern recognition

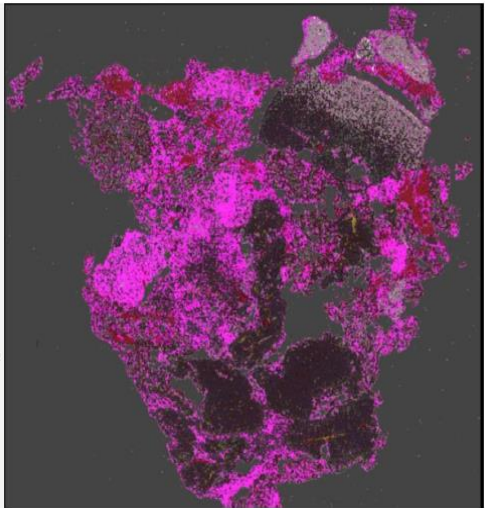
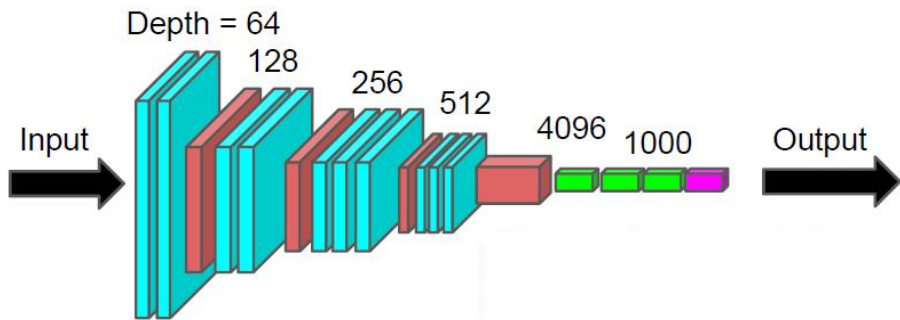
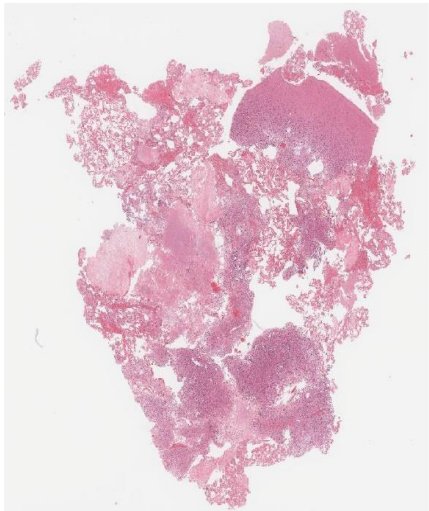


Step 1: Semantic segmentation

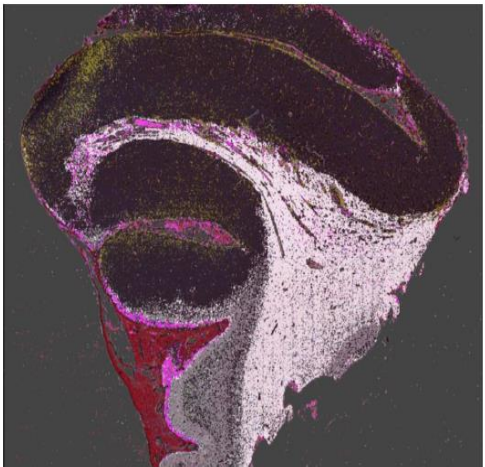
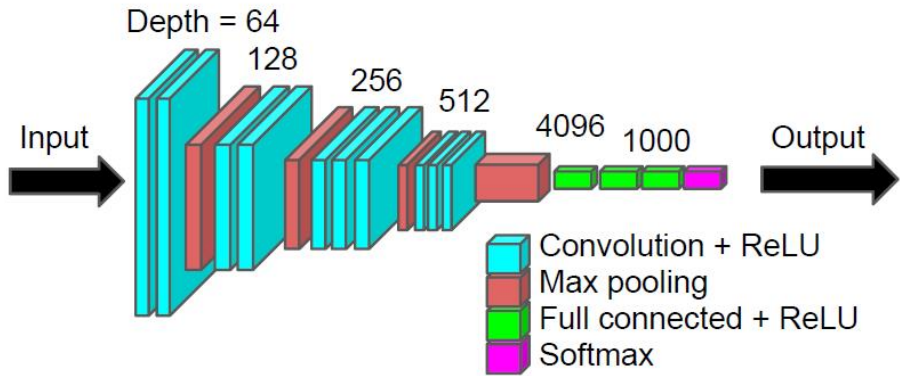
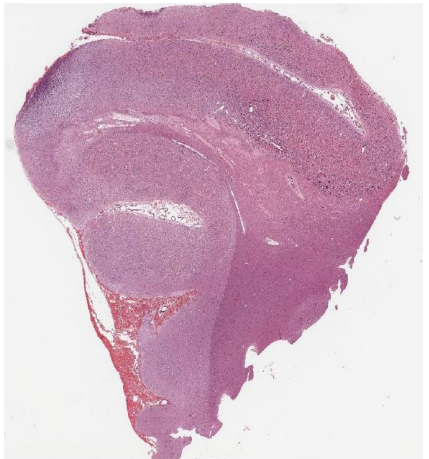
Neural Networks - Computer vision tools for automated pattern recognition



Quantitative semantic segmentation of cellular patterns



Necrosis	53.54%
Lesional Tissue	28.43%
Gray Matter	7.53%
Blood	6.44%
Surgical Material	3.87%



Lesional Tissue	55.06%
Gray Matter	24.38%
White Matter	24.38%
Blood	5.70%
Necrosis	3.34%

Deep learning-based classification of major brain tumors types

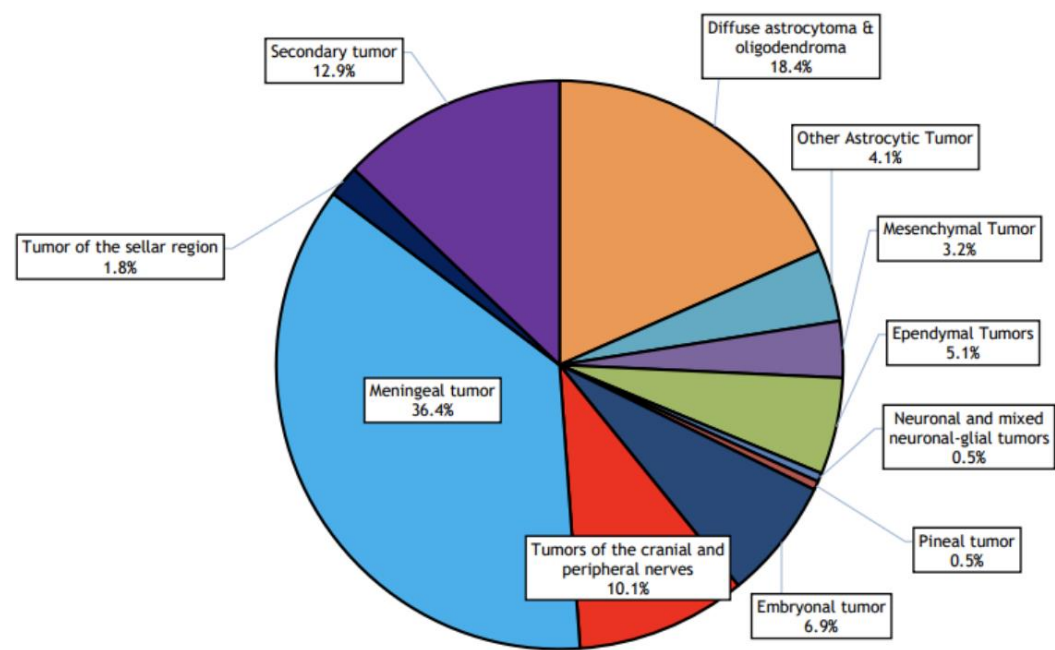
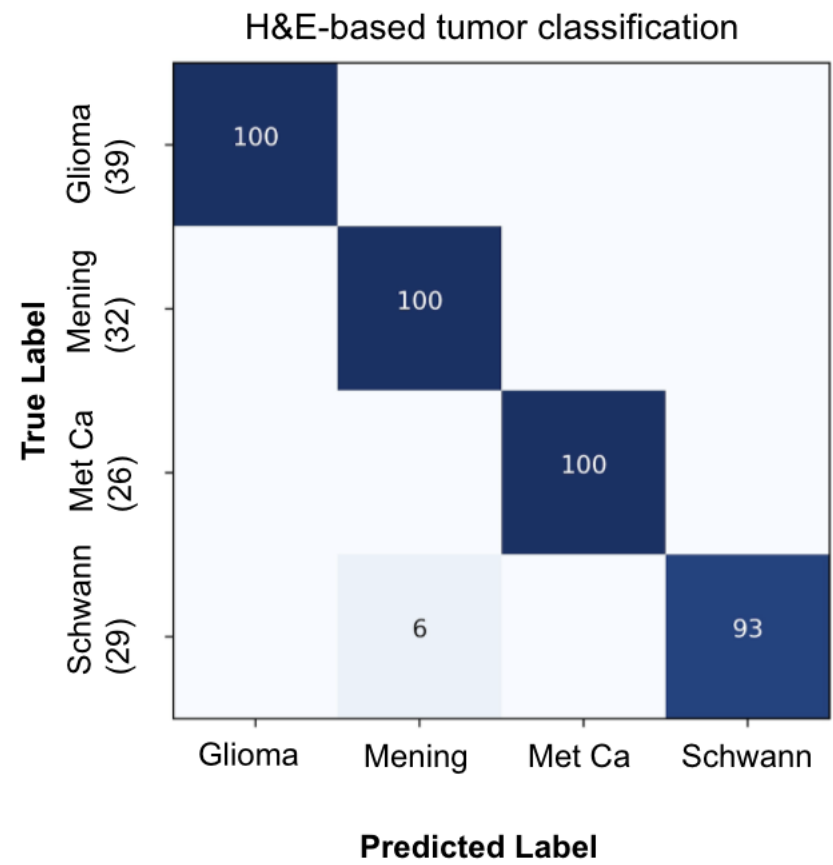


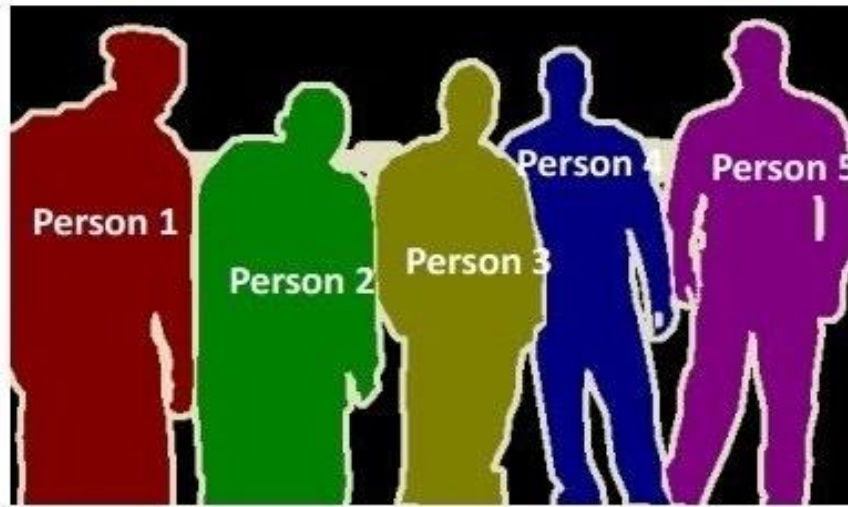
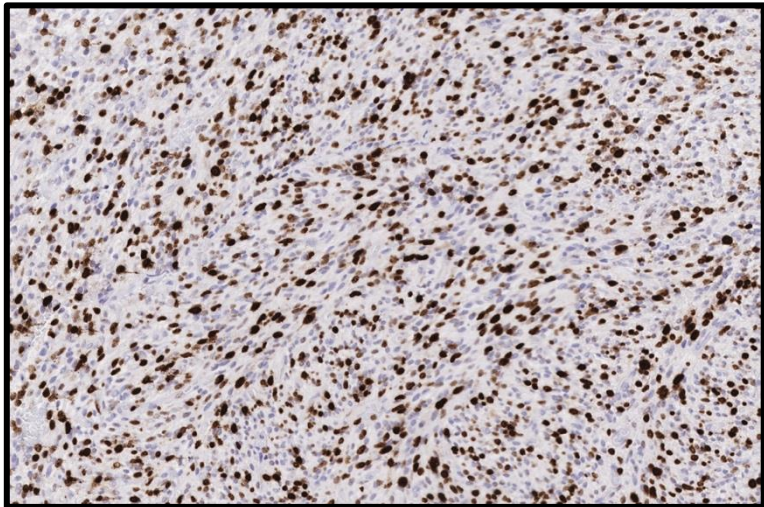
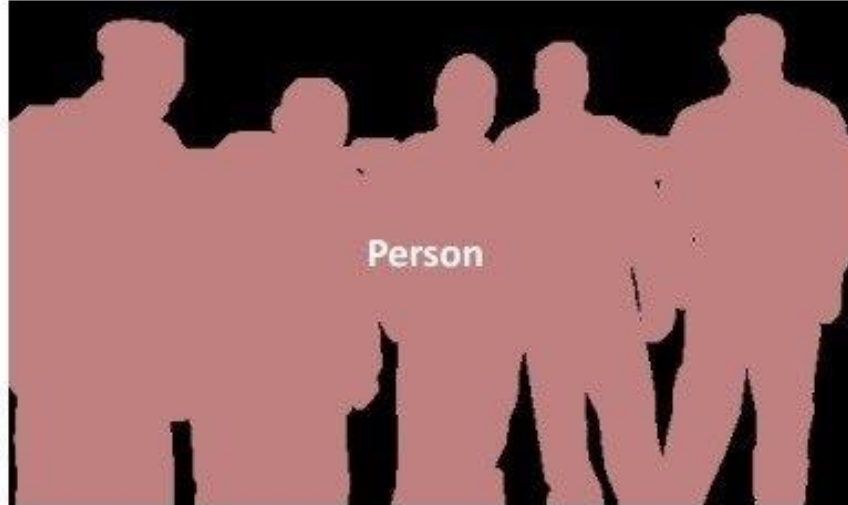
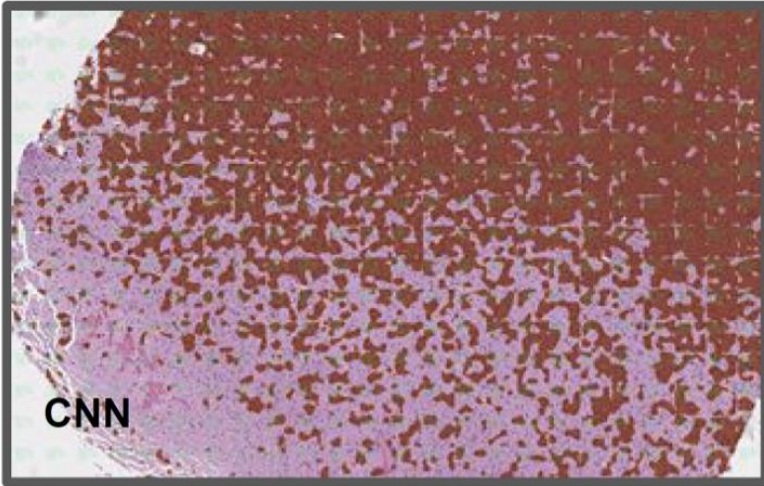
Figure 1: Distribution of major histological types of brain tumour.

Dzali et al, Mal Jour Pub Health Med 2017

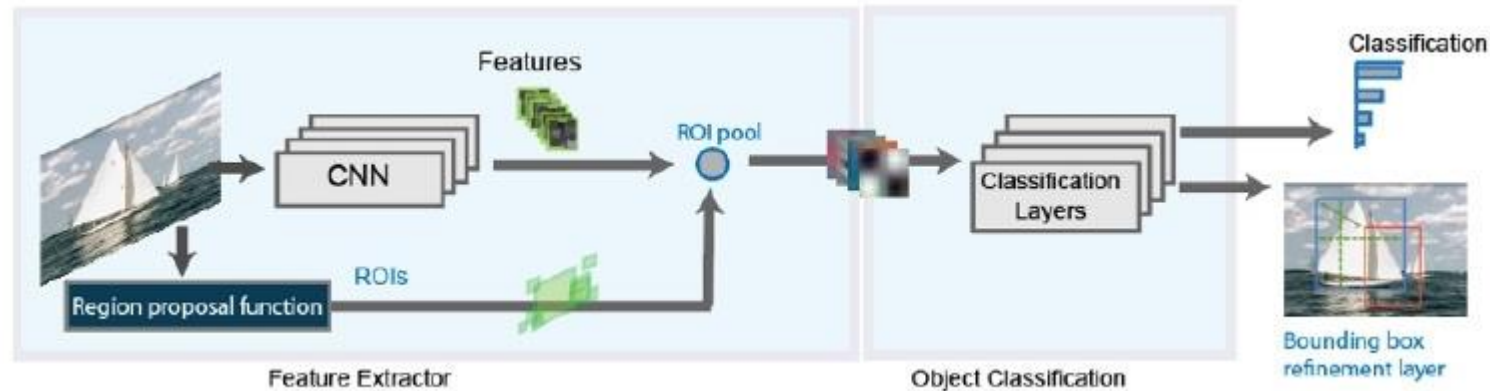
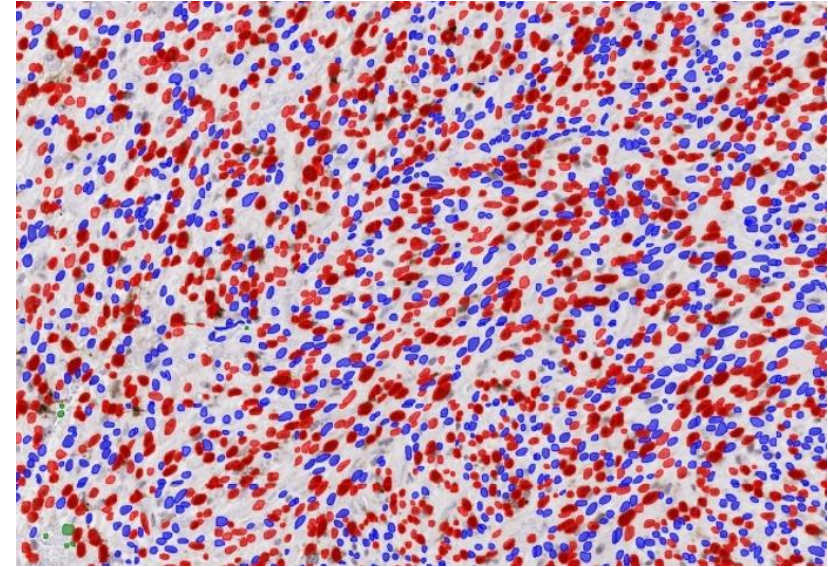
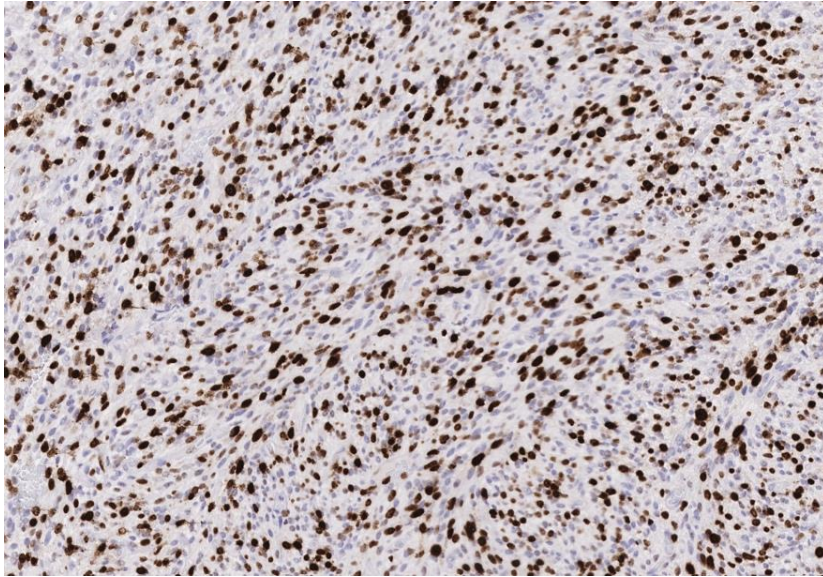


Step 2: Biomarker quantification (Instance segmentation)

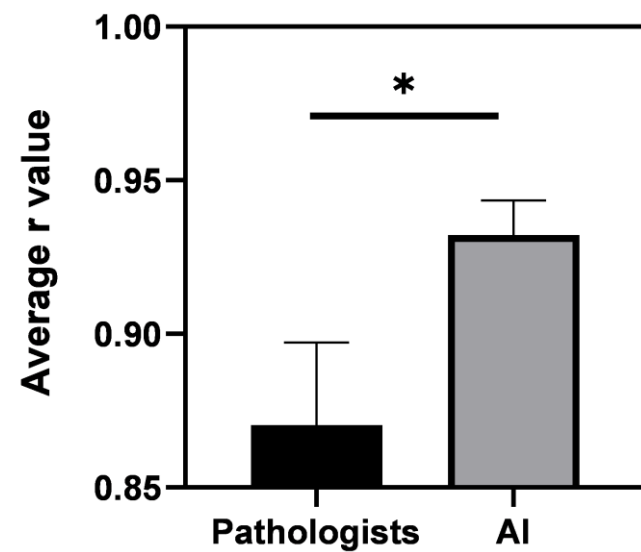
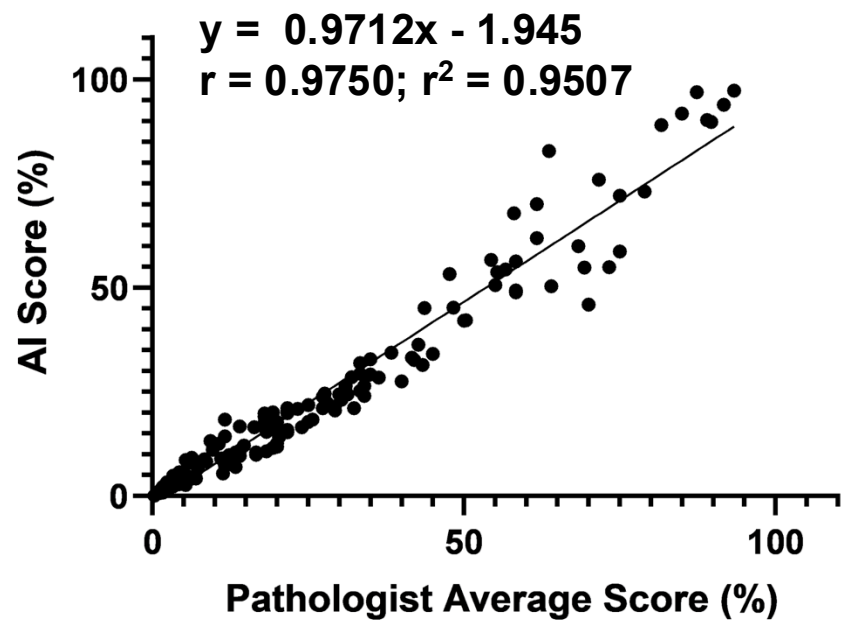
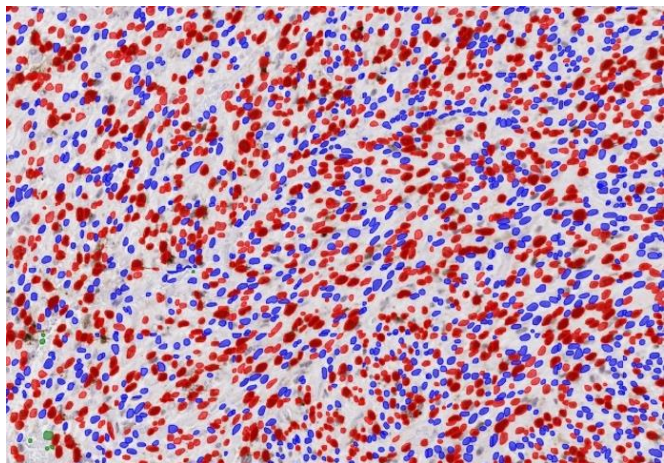
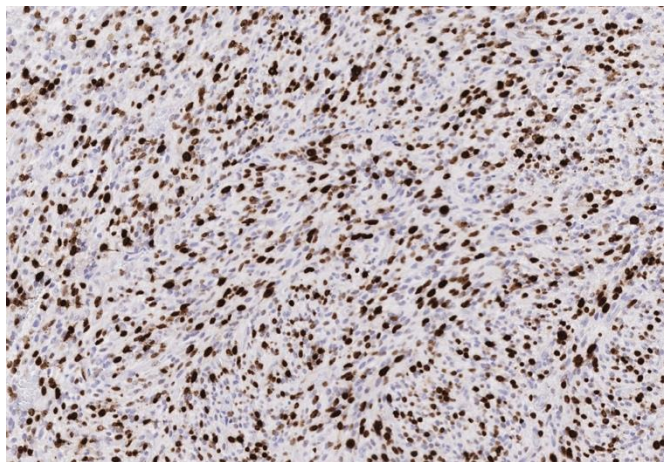
Mask R-CNN: Instance segmentation = Semantic segmentation + object detection



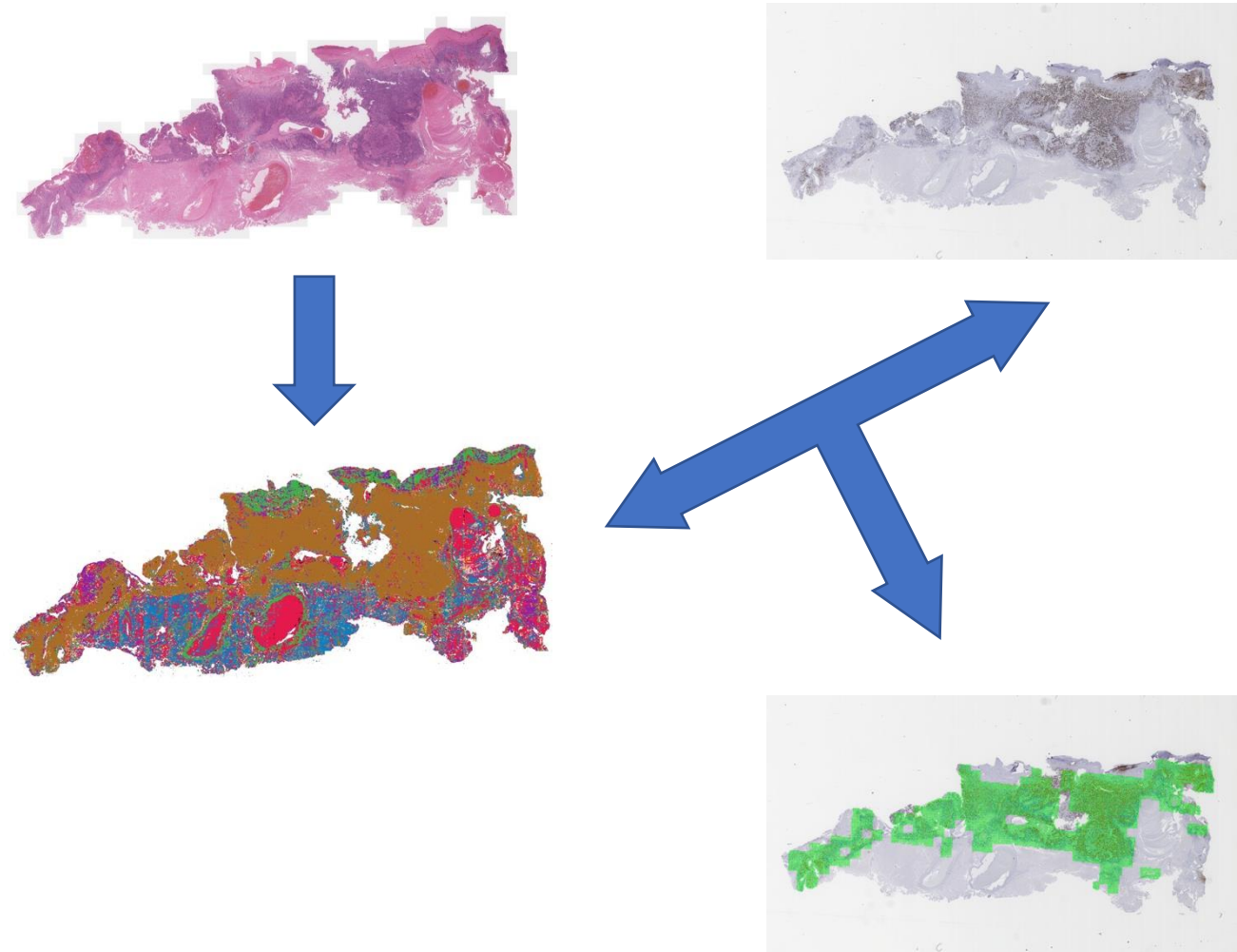
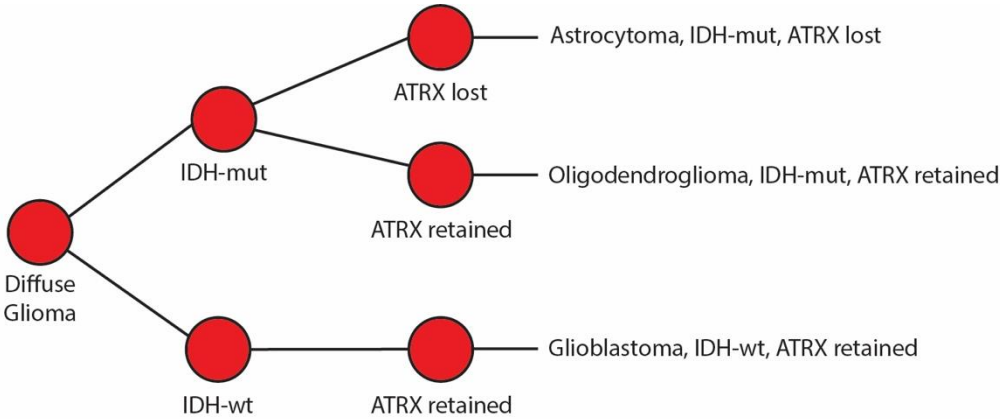
Nuclear segmentation and quantification with Mask R-CNN



Cell quantification with Mask R-CNN

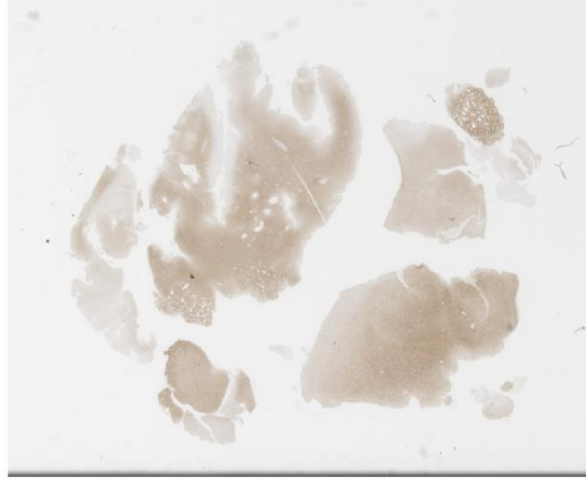
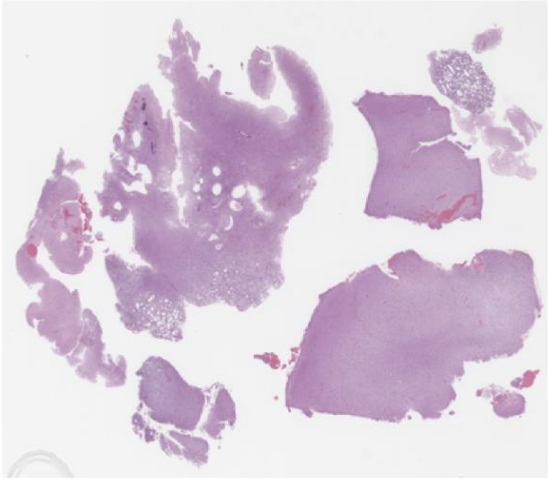


Step 3: Biomarker Interpretation (Image registration)

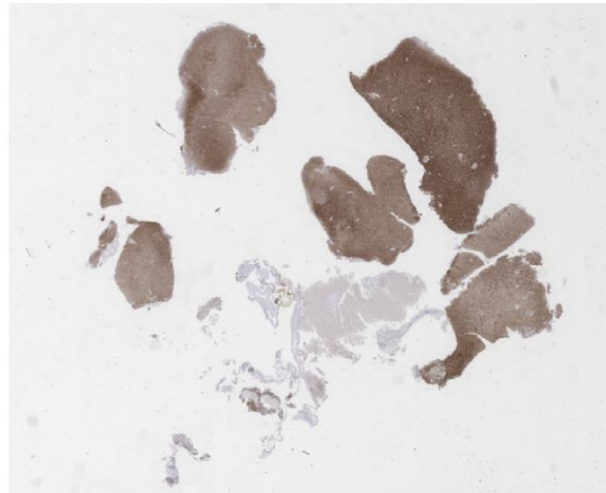
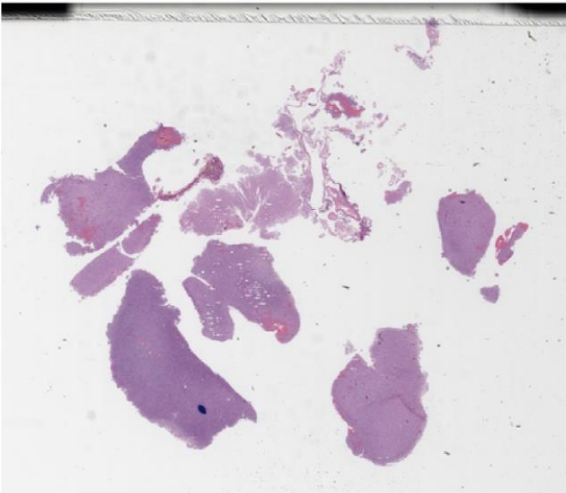


Challenges: Translation and Rotation in Tissue

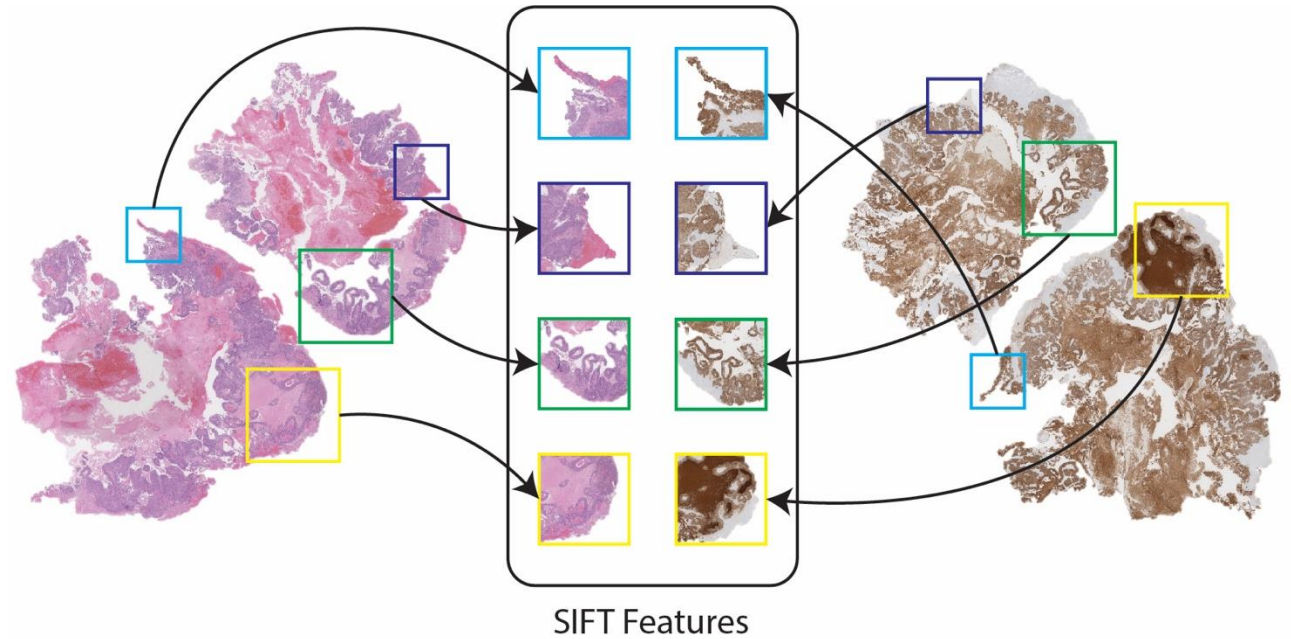
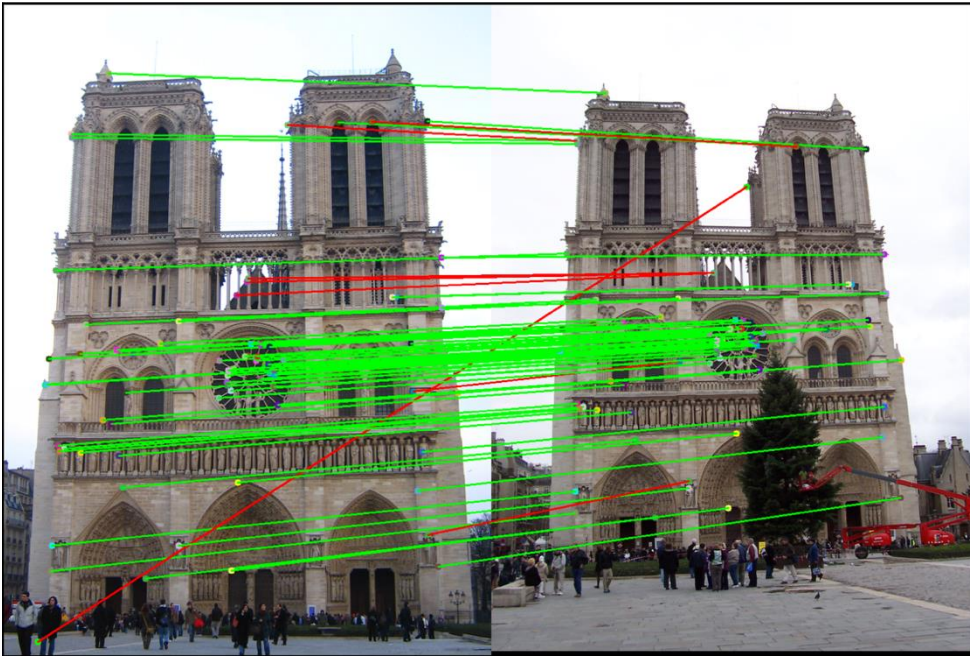
Horizontal
Translation



180°
Rotation

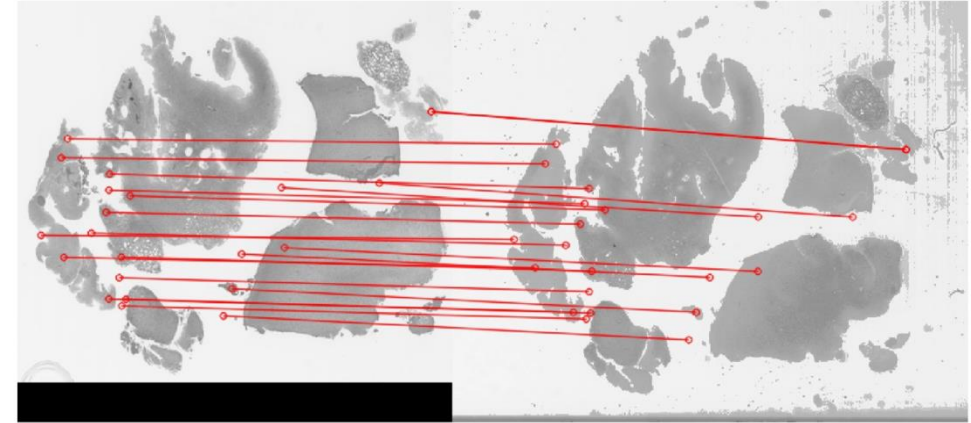
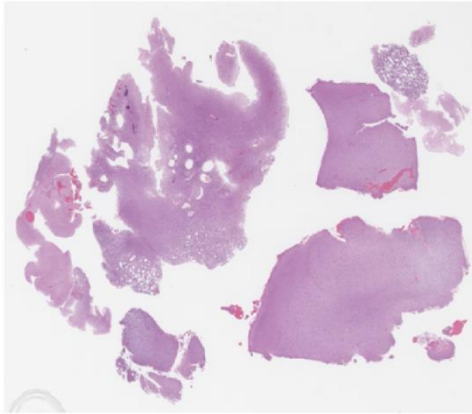


Computer vision solution: Image matching / Image Registration

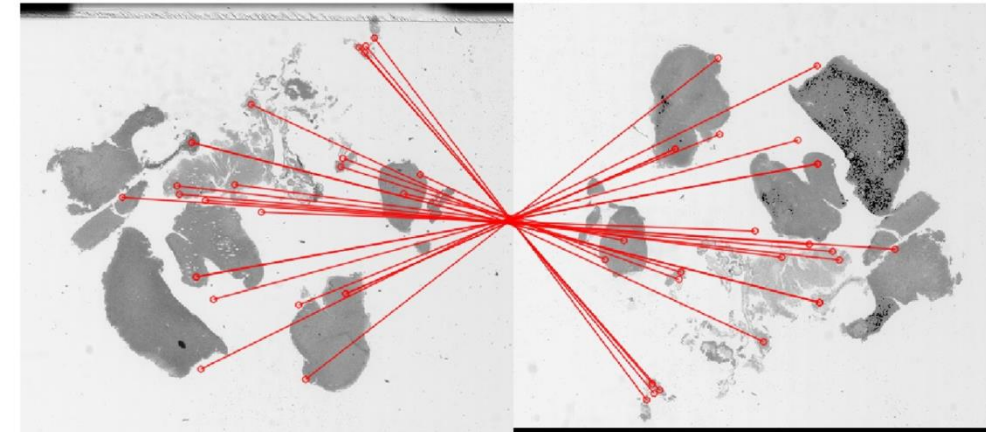
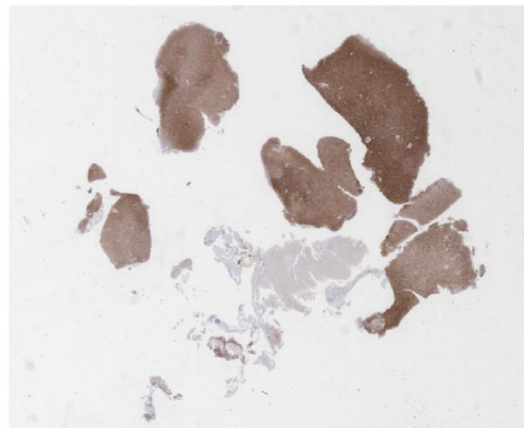
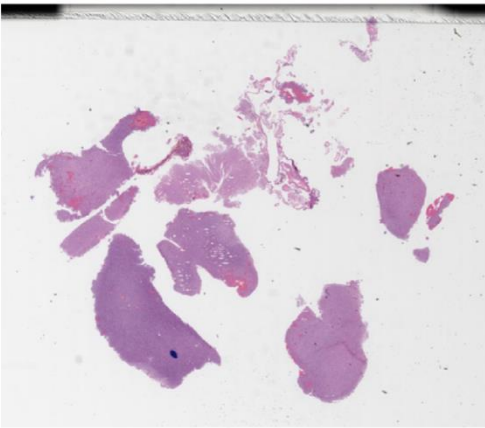


Automated WSI registration using computer vision

Horizontal
Translation

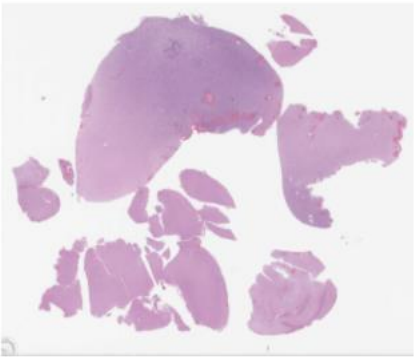


180°
Rotation

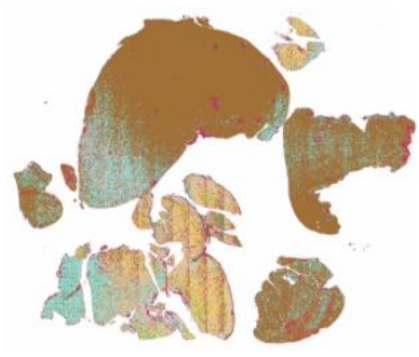


Applications: Automating molecular subclassification of brain tumors

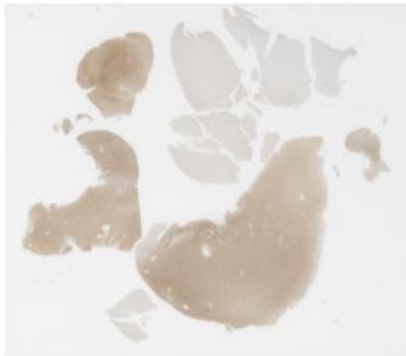
H&E



CNN H&E



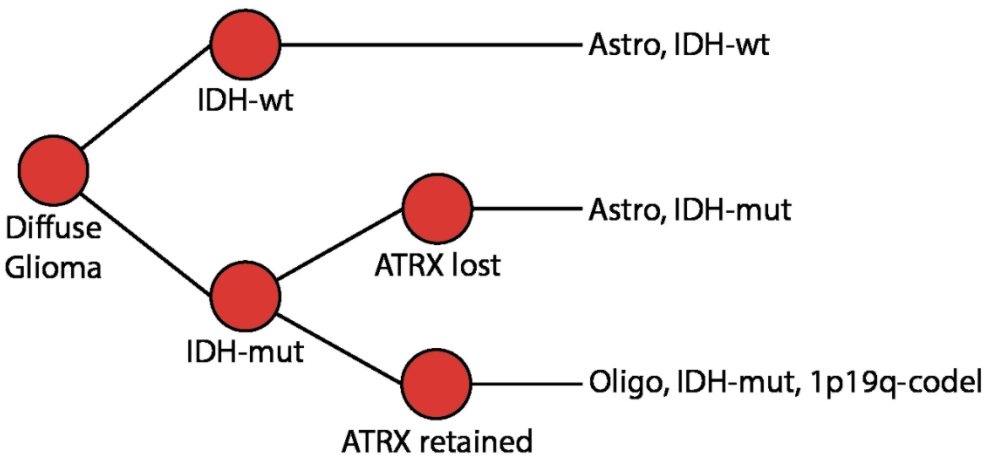
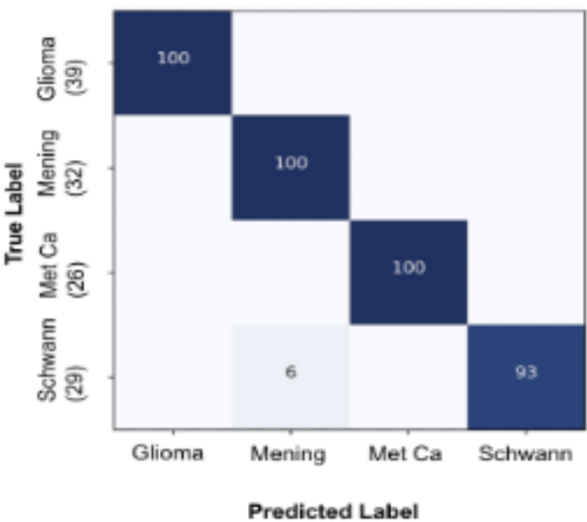
IHC



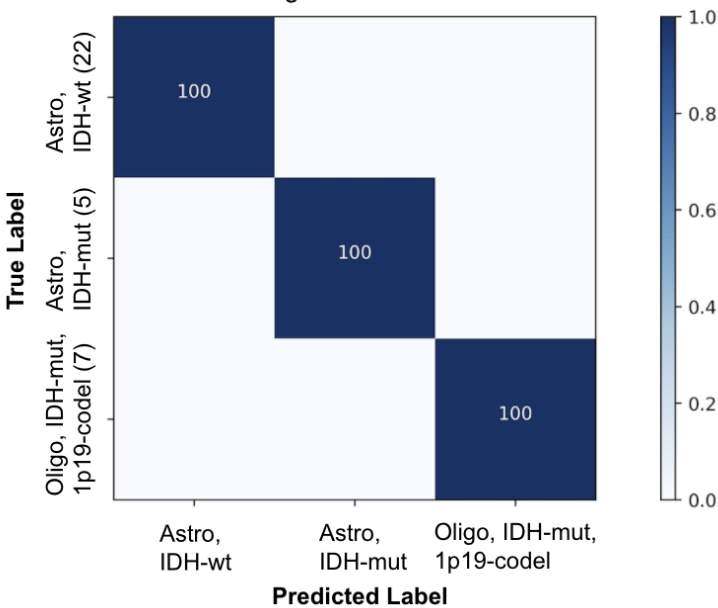
SIFT IHC



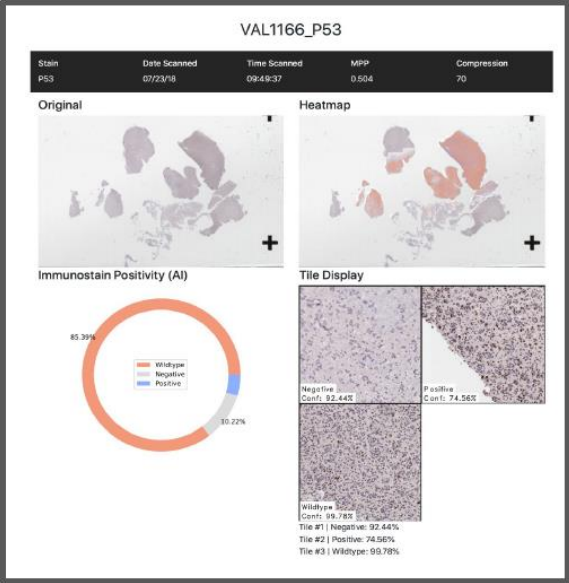
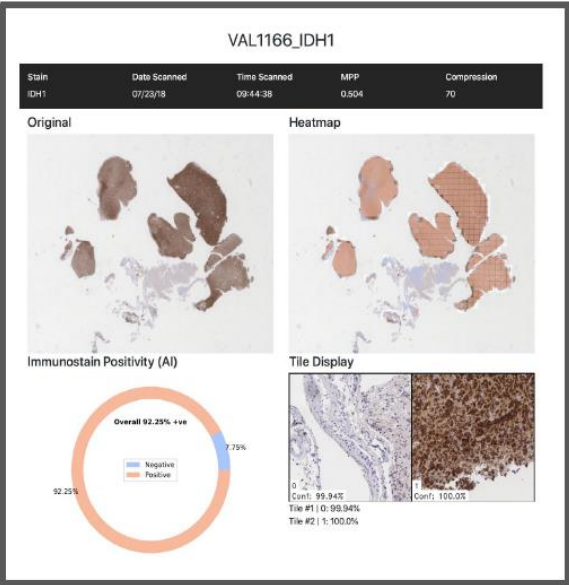
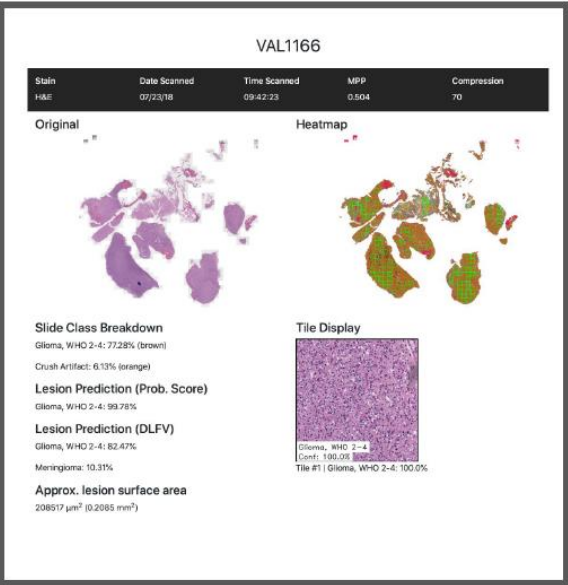
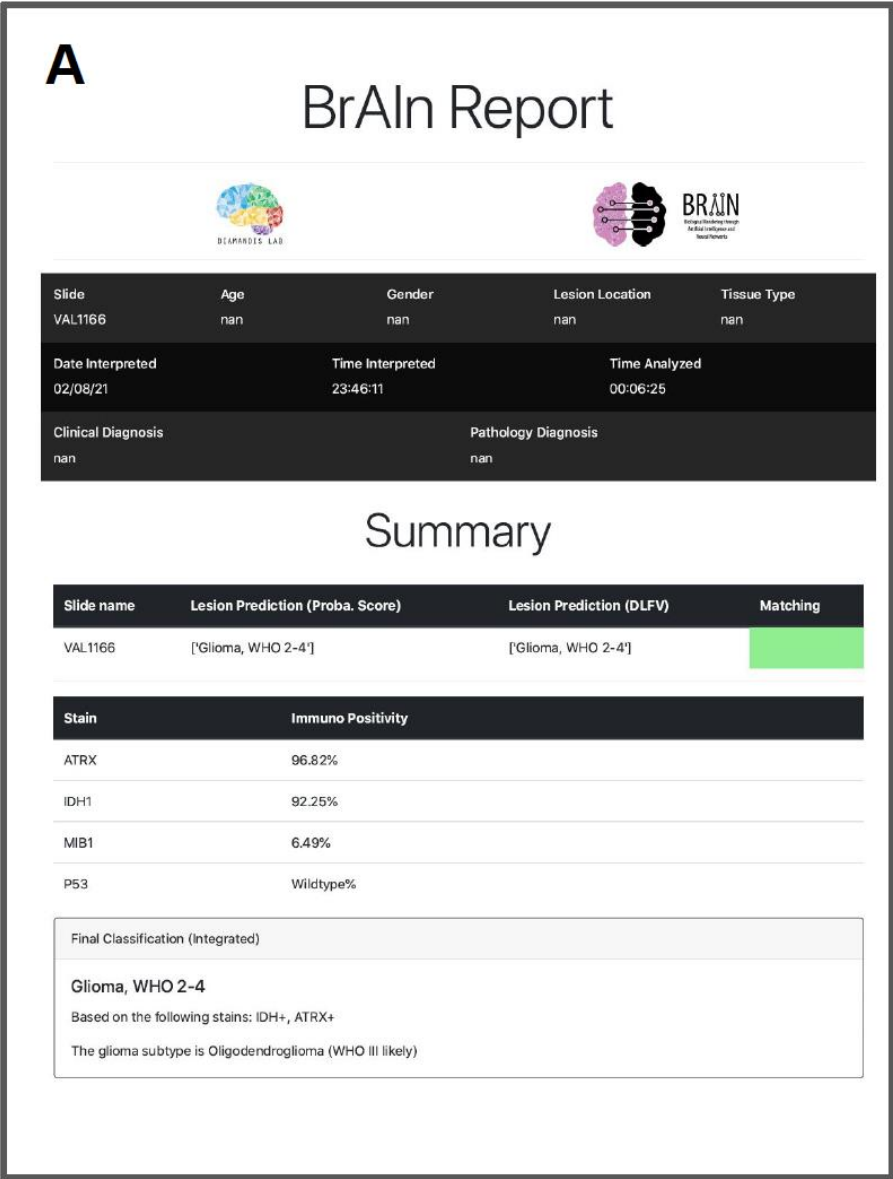
H&E-based tumor classification



IHC-based glioma subclassification



Integrating molecular information into pathology reports



Integrating molecular information into pathology reports

B

BrAIn Report





Slide	Age	Gender	Lesion Location	Tissue Type
VAL1969	nan	nan	nan	nan
Date Interpreted 02/08/21		Time Interpreted 23:13:49		Time Analyzed 00:08:41
Clinical Diagnosis nan		Pathology Diagnosis nan		

Summary

Slide name	Lesion Prediction (Proba. Score)	Lesion Prediction (DLFV)	Matching
VAL1969	['Glioma, WHO 2-4']	['Glioma, WHO 2-4']	

Stain	Immuno Positivity
ATRX	8.45%
IDH1	92.57%
MIB1	18.18%
P53	Positive%

Final Classification (Integrated)

Glioma, WHO 2-4

Based on the following stains: IDH+, ATRX-

The glioma subtype is GBM Astrocytoma (WHO IV likely)

VAL1969

Stain	Date Scanned	Time Scanned	MPP	Compression
H&E	11/23/18	08:37:38	0.504	70

Original

Heatmap

Slide Class Breakdown

Glioma, WHO 2-4: 90.86% (brown)

Radiation Necrosis: 5.17% (grey)

Lesion Prediction (Prob. Score)

Glioma, WHO 2-4: 99.04%

Lesion Prediction (DLFV)

Glioma, WHO 2-4: 67.94%

Meningioma: 25.90%

Approx. lesion surface area

596786 μm^2 (0.5968 mm^2)

Tile Display

Glioma, WHO 2-4

Conf: 100.00%

Tile #1 | Glioma, WHO 2-4: 100.00%

VAL1969_IDH1

Stain	Date Scanned	Time Scanned	MPP	Compression
IDH1	11/23/18	08:39:59	0.504	70

Original

Heatmap

Immunostain Positivity (AI)

Overall 92.57% +ve

92.57%

4.3%

Tile Display

Conf: 99.94%

Tile #1 | 99.99%

Tile #2 | 100.0%

VAL1969_ATRX

Stain	Date Scanned	Time Scanned	MPP	Compression
ATRX	11/23/18	08:43:06	0.504	70

Original

Heatmap

Immunostain Positivity (AI)

Overall 8.45% +ve

91.55%

8.45%

Tile Display

Conf: 100.00%

Tile #1 | 100.0%

Tile #2 | 99.94%

VAL1969_P53

Stain	Date Scanned	Time Scanned	MPP	Compression
P53	11/23/18	08:40:50	0.504	70

Original

Heatmap

Immunostain Positivity (AI)

Overall 99.72% +ve

99.72%

Tile Display

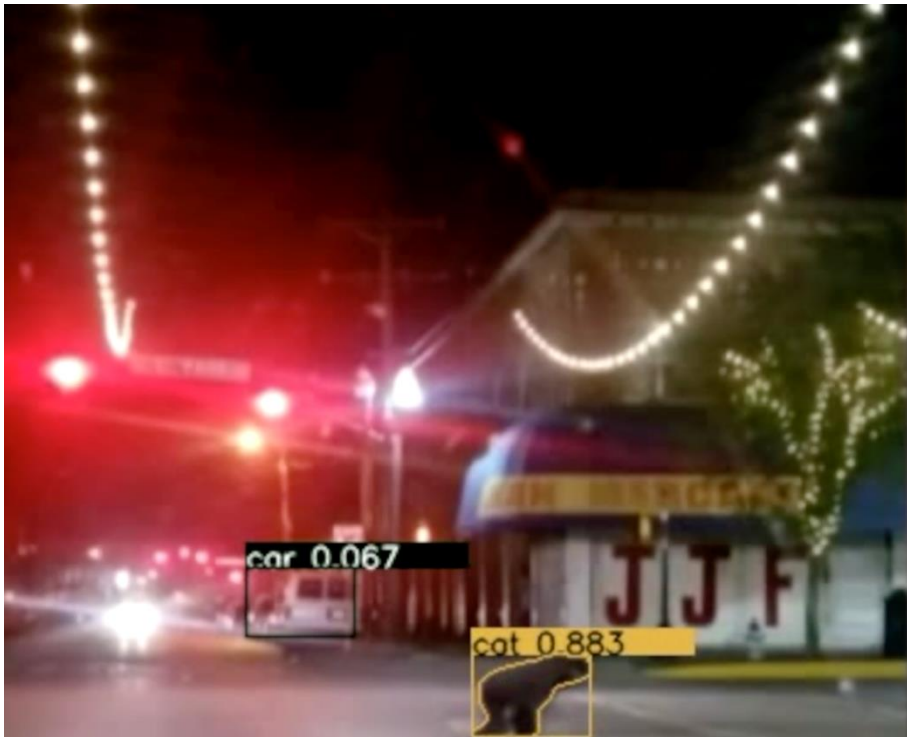
Conf: 100.00%

Tile #1 | Positive: 100.0%

Implementation Challenges

“Partial Solutions” - Too many “edge cases”

Deep learning + Data = Not enough



Edge Cases

- “Edges”: All the things that can go wrong in real world that you never expected
- Everyone logging into your site at the same time
- Hard to catalog: Individually rare, but can be collectively a common problem
- Autonomous driving: To accumulate enough rare incidents (e.g. crashes, close calls) to train vehicles, you need billions of miles → would require hundreds of years to develop training sets



Edge Cases

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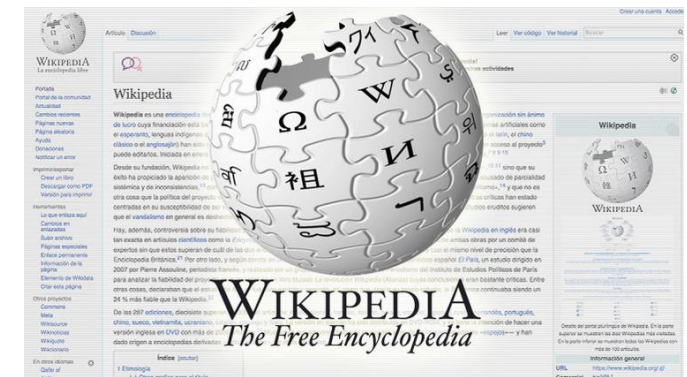
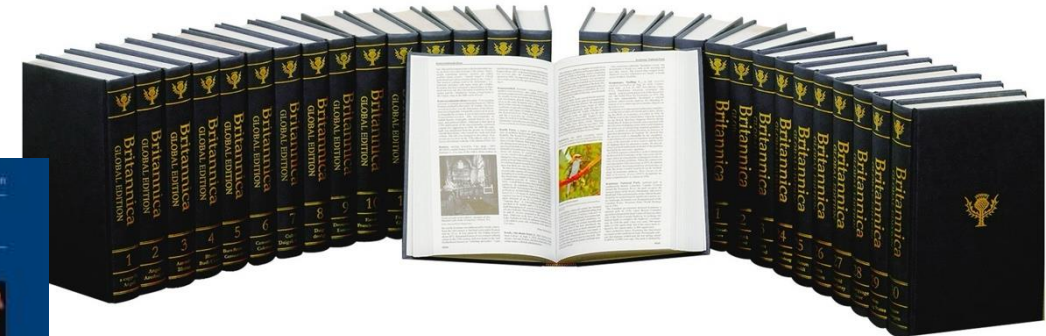
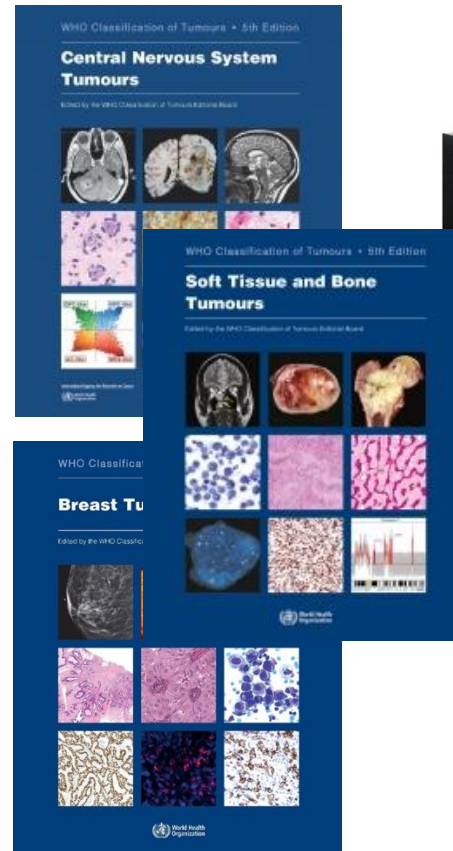


Implementation Challenges

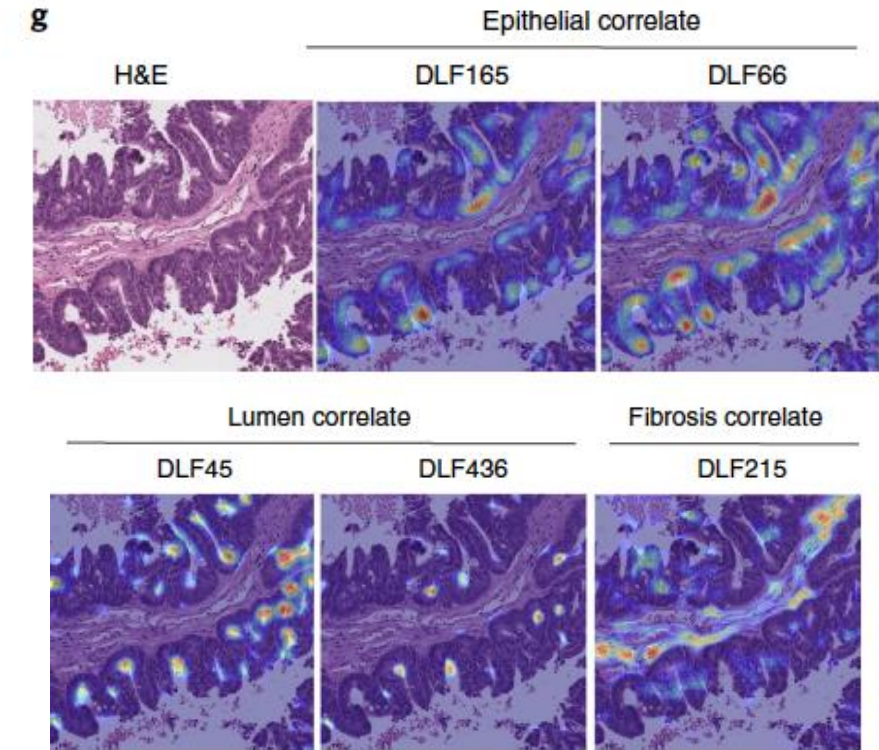
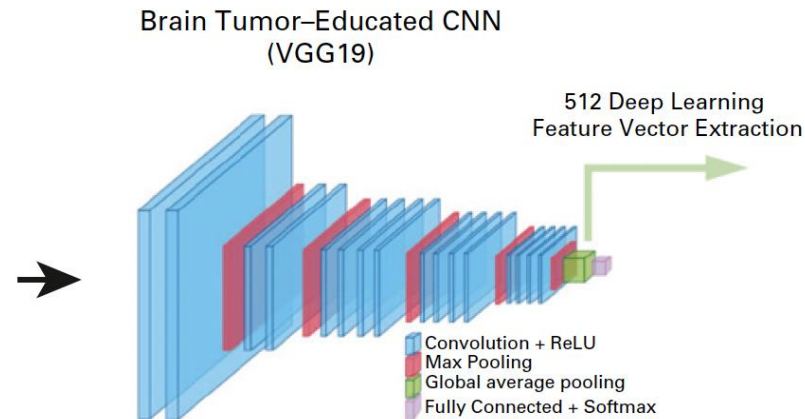
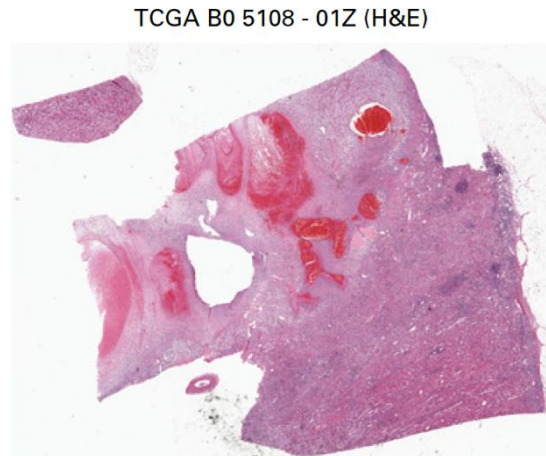
“Partial Solutions” - Too many “edge cases”

Deep learning + Data = Not enough

- Each neural network designed for a single highly specialize task
- Each pathology subspecialty has 100's of specialized tasks
- Multiple subspecialties
- Solution:
 - Crowdsourcing
 - Computational pathology

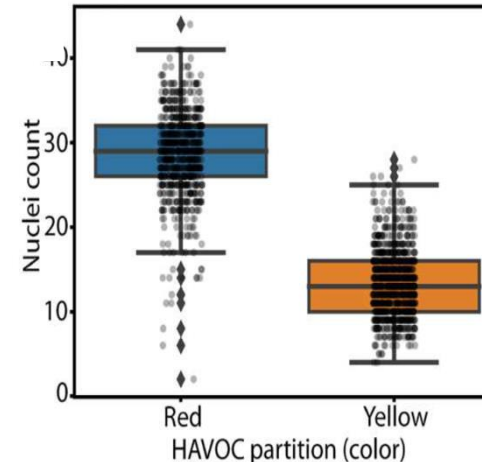
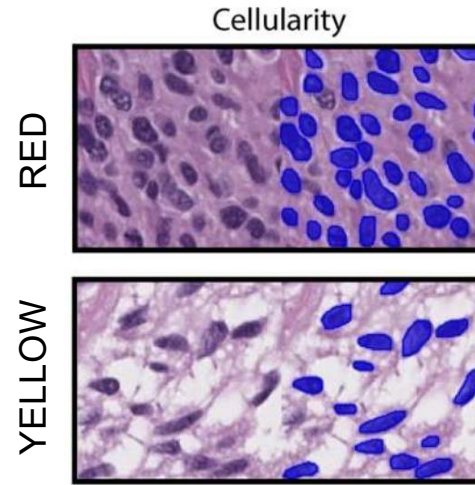
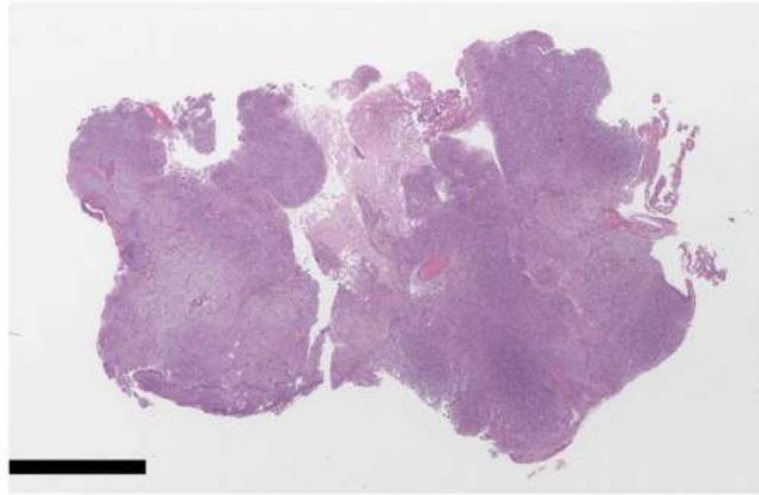


Deep Feature Extraction – Unsupervised analysis

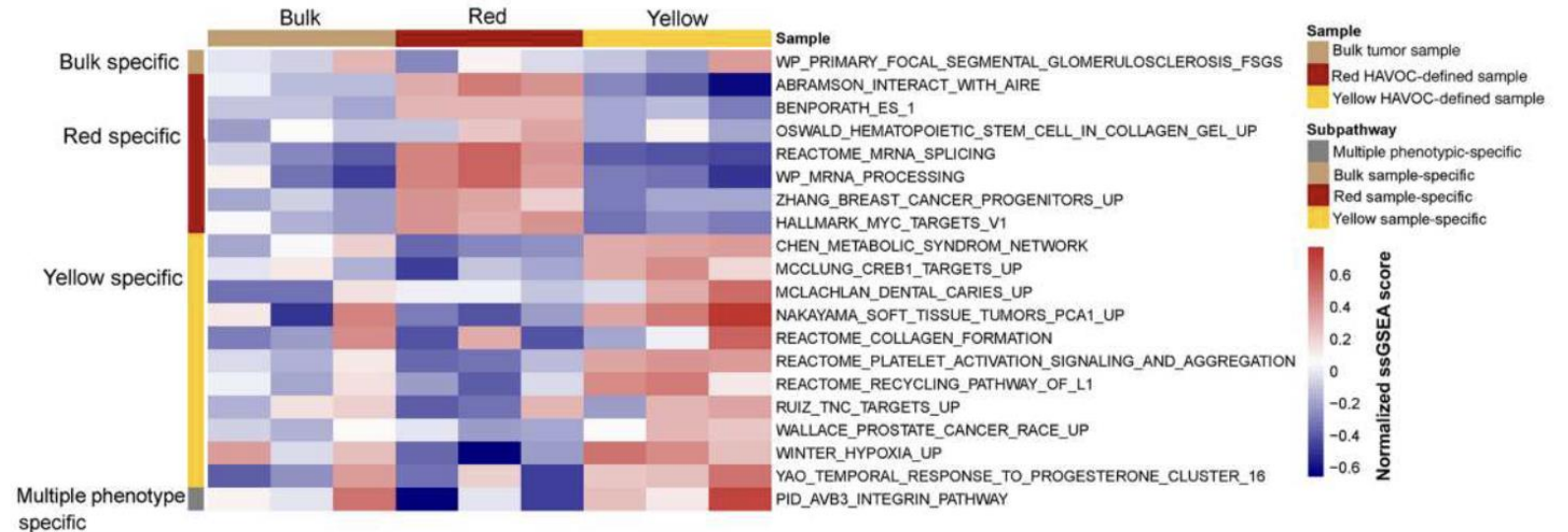
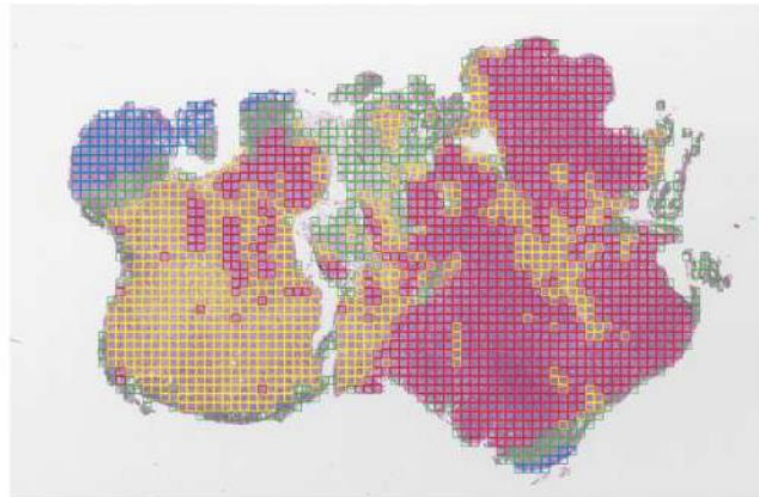
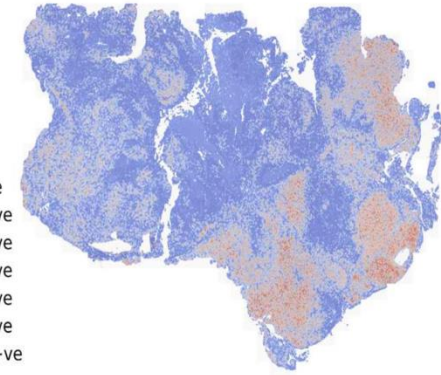
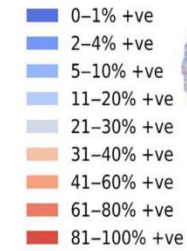


Faust et al, *Nature Machine Intelligence*, 2019

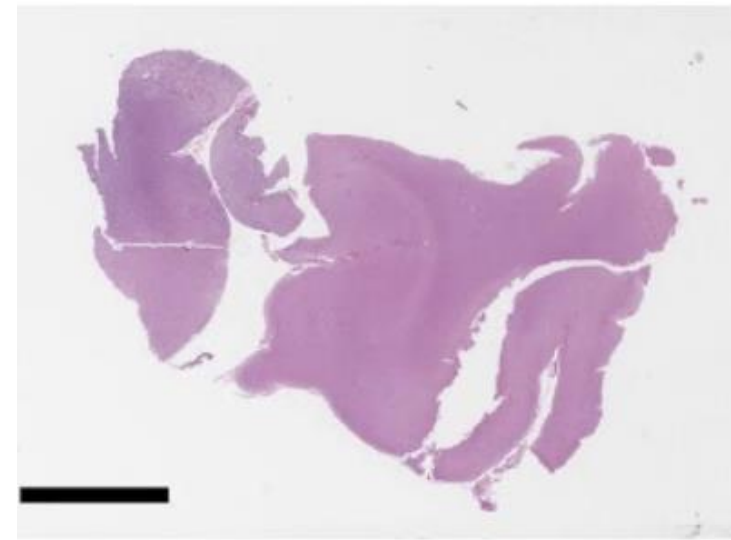
Automating analysis of Intra-Tumor Heterogeneity



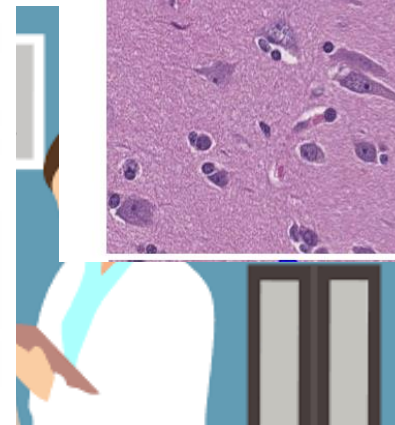
Ki-67



PHARAOH: Weakly supervised Active Learning + Crowd Sourcing

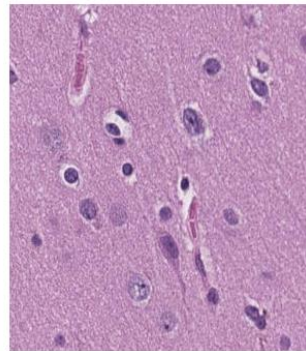


$\mathcal{M}$

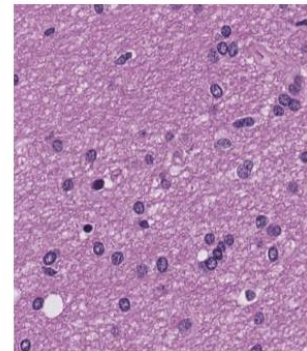


Trained Classifiers

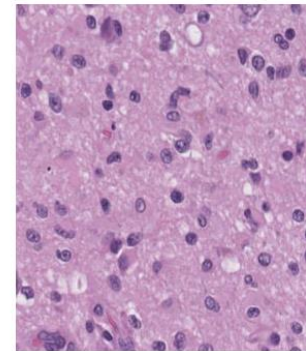
Yellow



Green

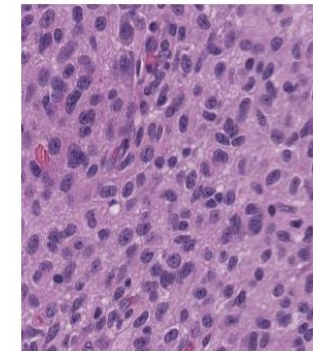


Purple

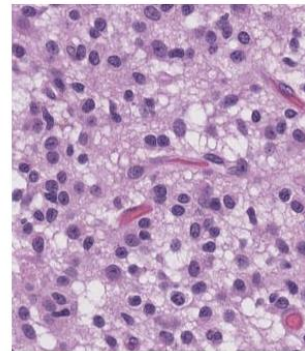
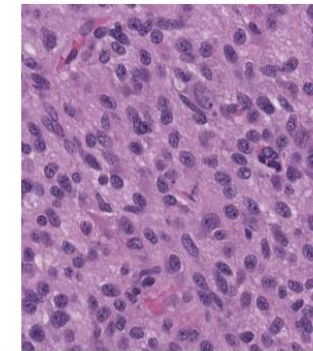
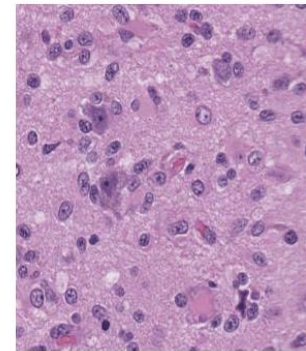
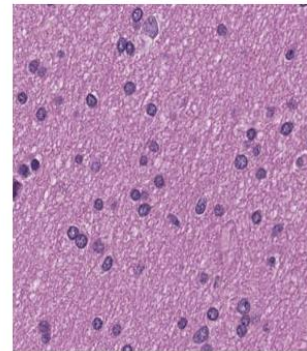
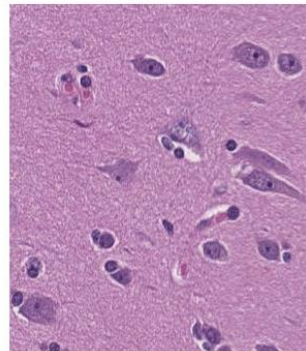
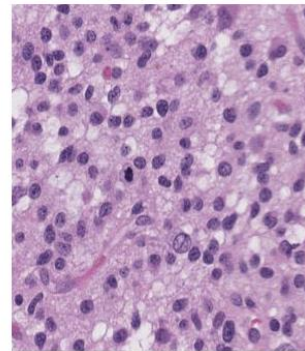


Brain Classifier

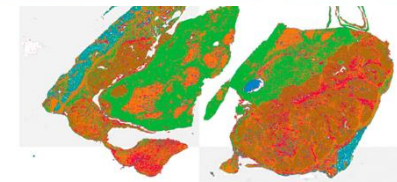
Cyan



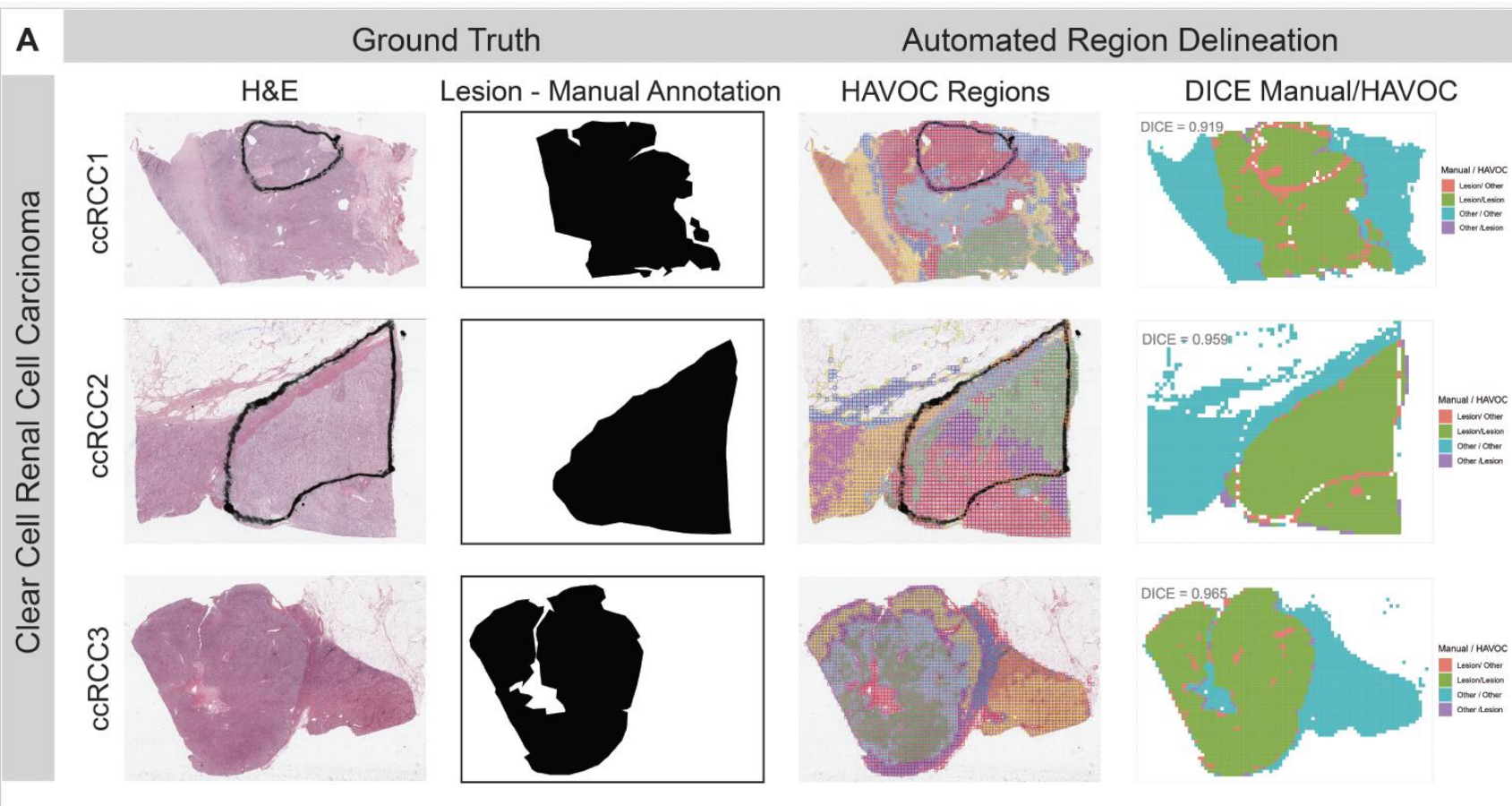
Blue



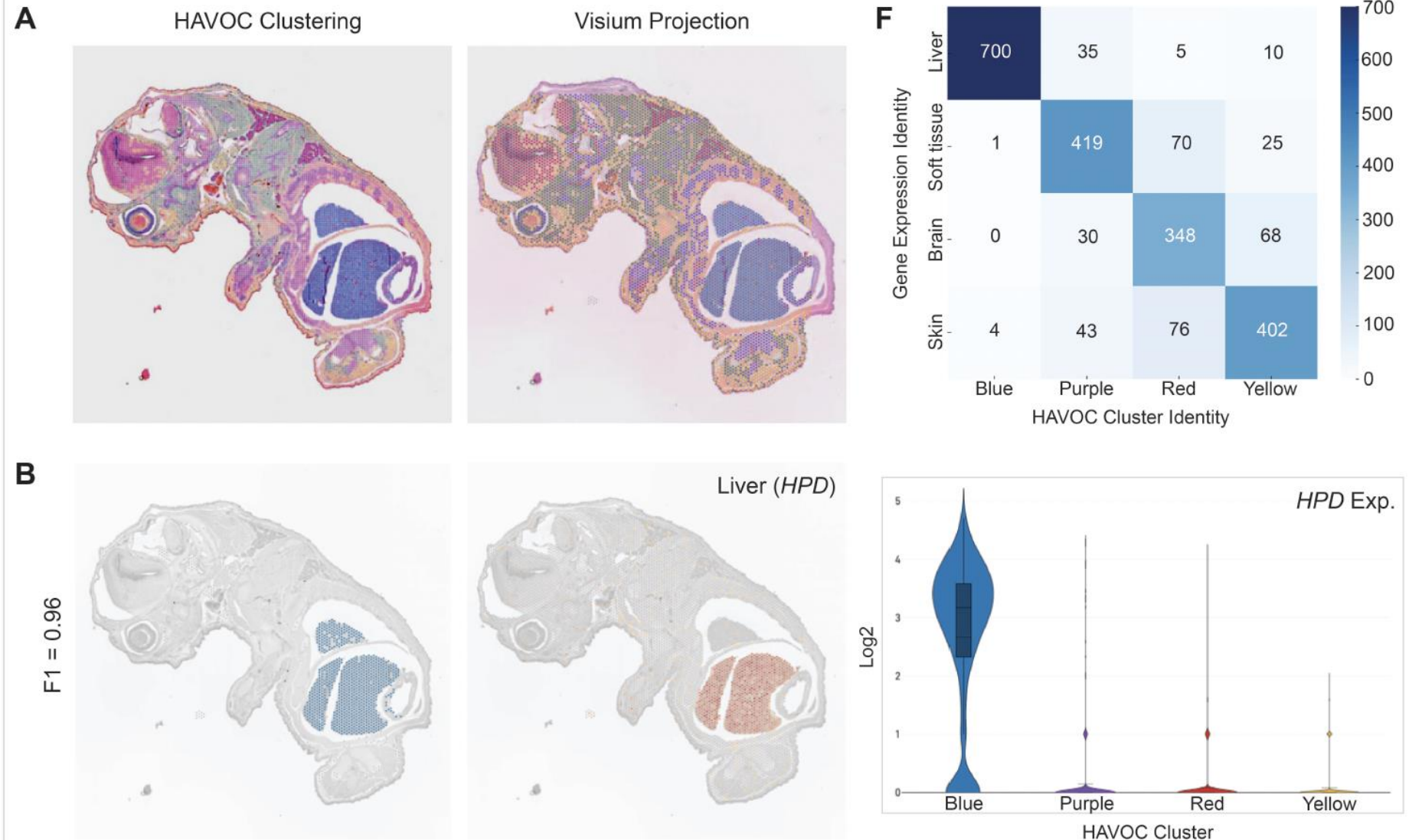
Pattern Inferences



PHARAOH: Automated PHARAOH Segmentation and Developed Classifiers align with Expert and molecular ground truths



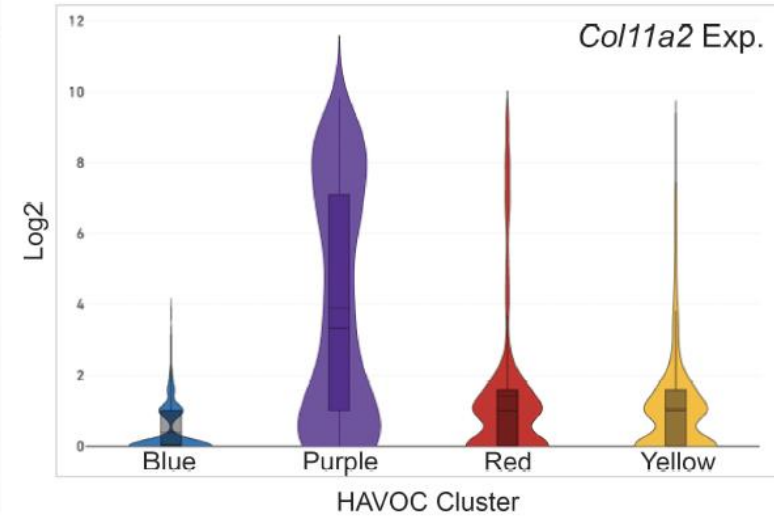
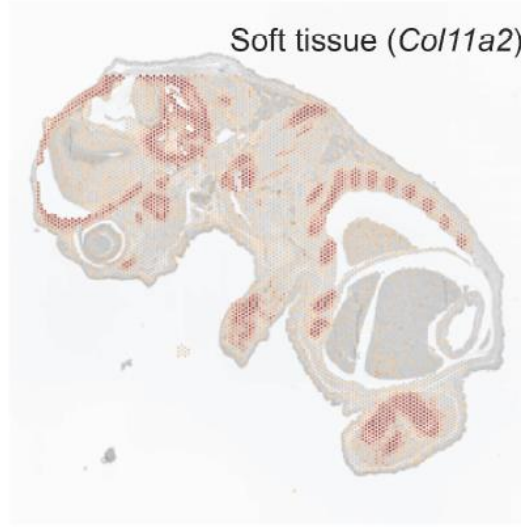
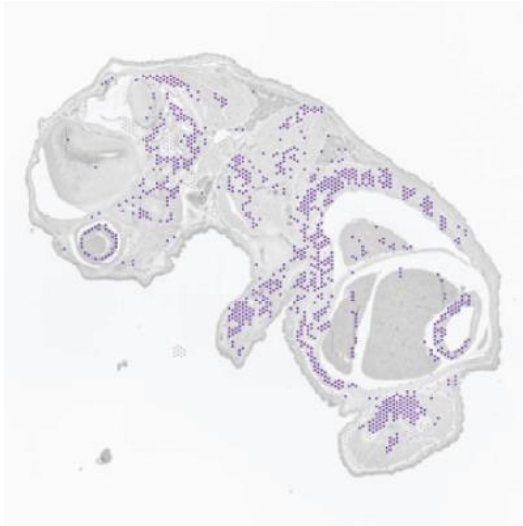
PHARAOH: Automated PHARAOH Segmentation and Developed Classifiers align with Expert and molecular ground truths



PHARAOH: Automated PHARAOH Segmentation and Developed Classifiers align with Expert and molecular ground truths

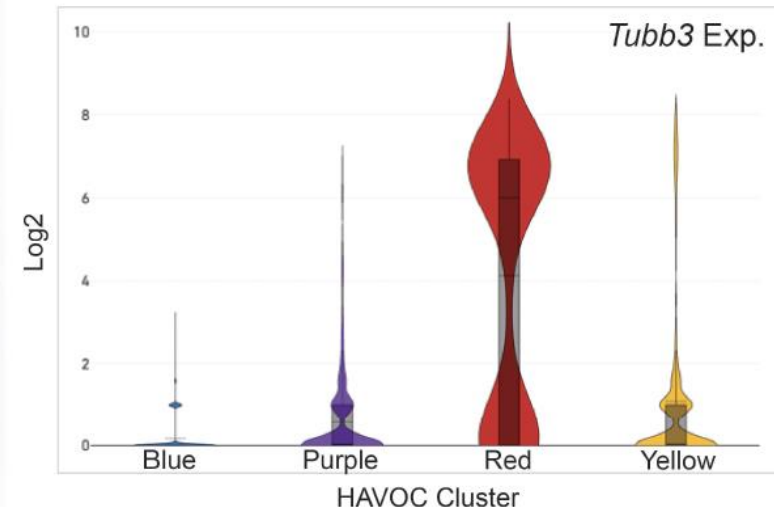
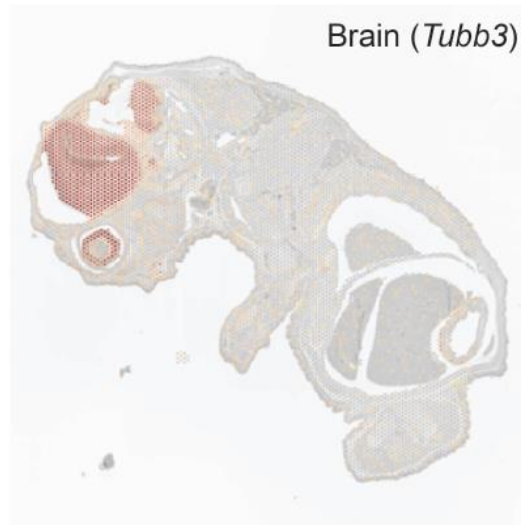
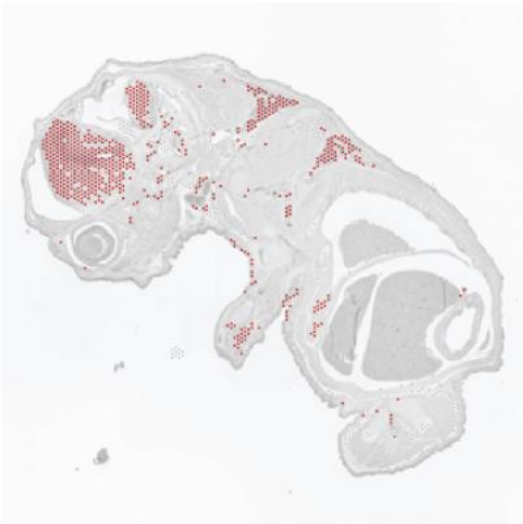
C

F1 = 0.80

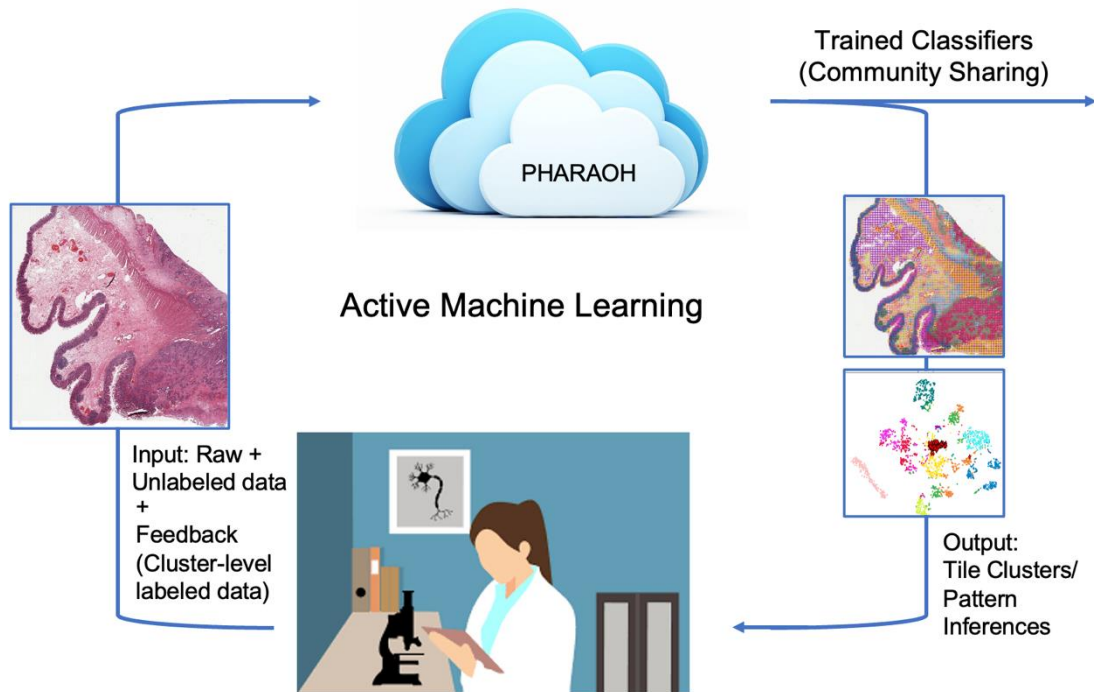


D

F1 = 0.74

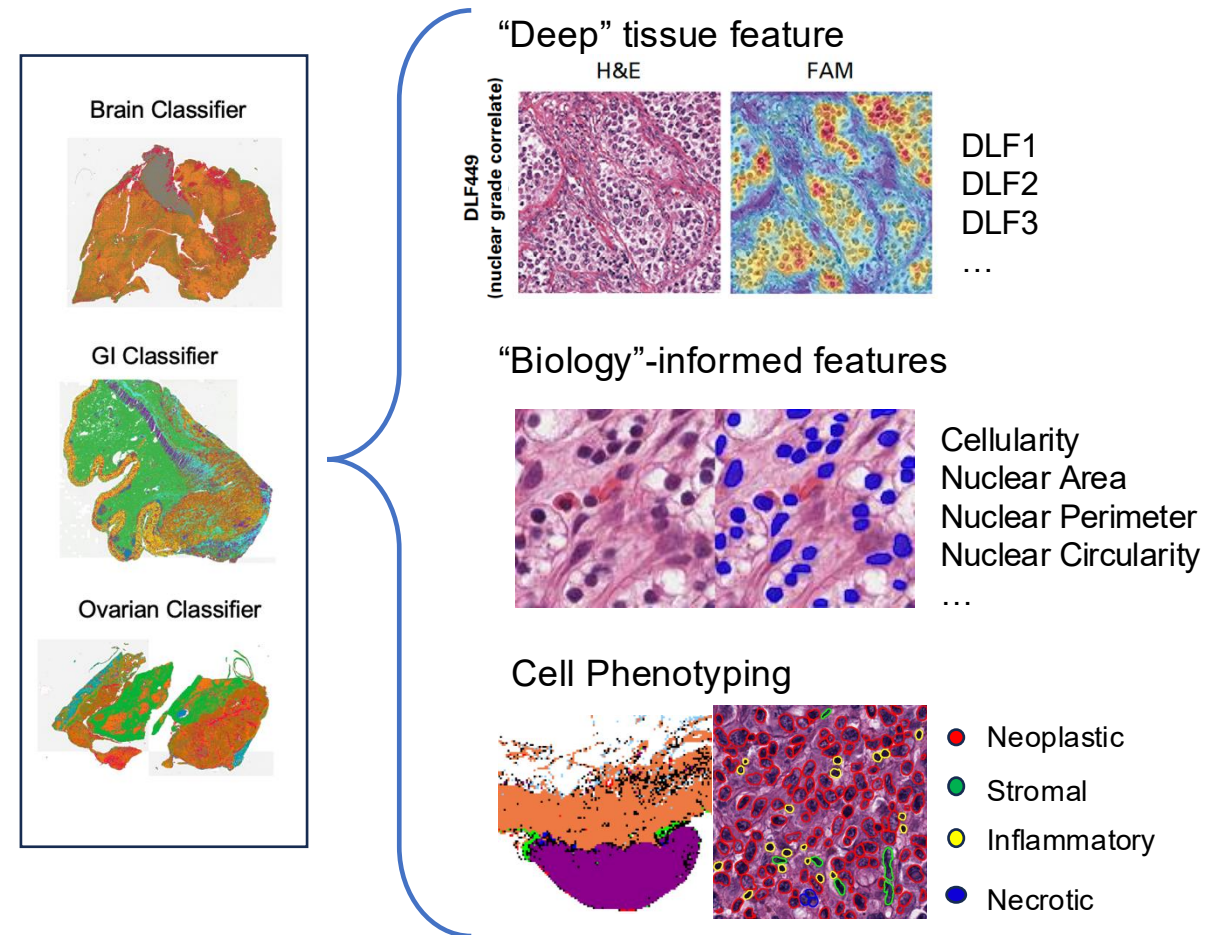


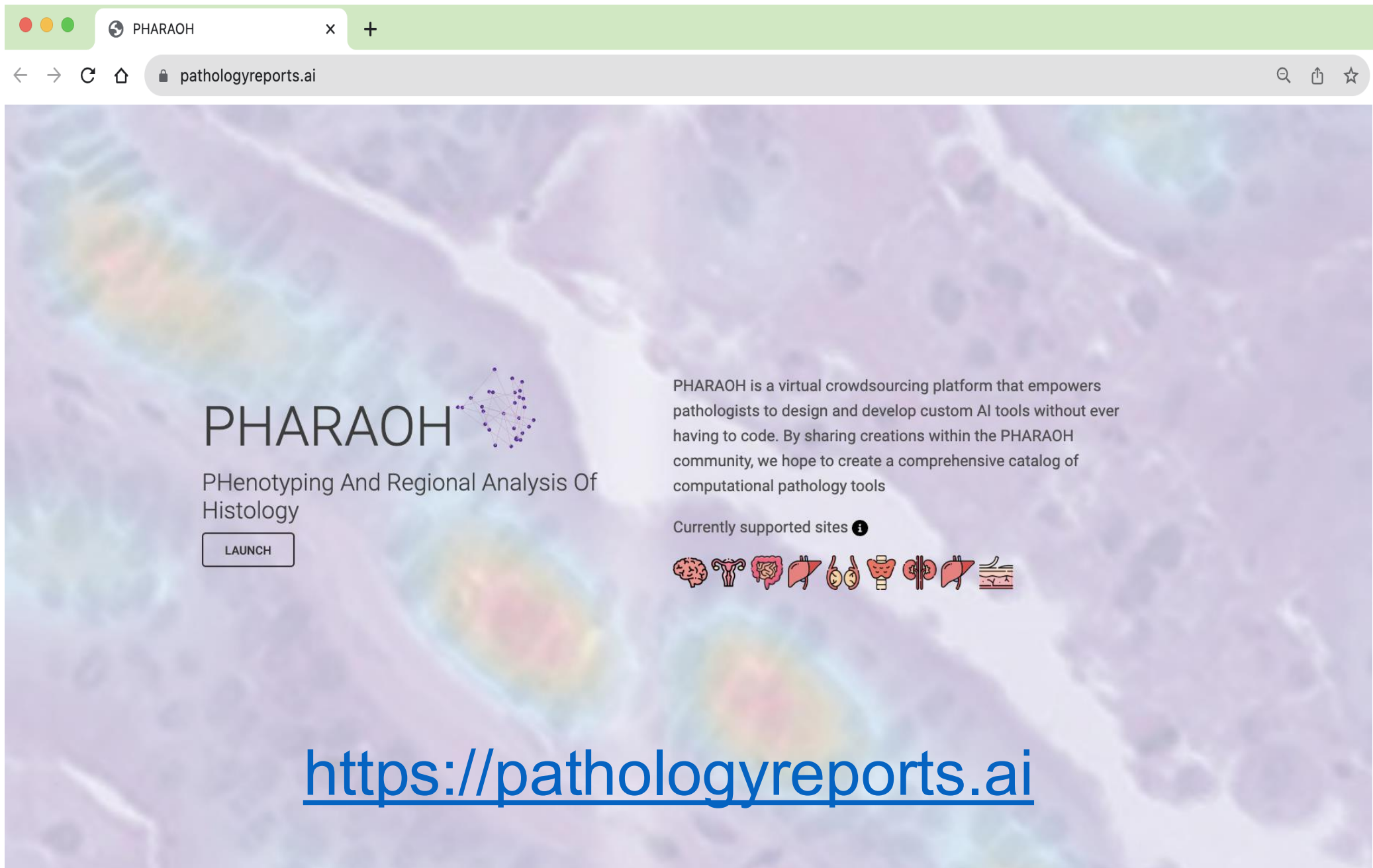
Crowdsourcing Classifier Production



Biomarker (Feature) Extraction

Outputs: Phenotyping and Regional Analysis of Histology





PHARAOH



PHeNOTyping And Regional Analysis Of
Histology

LAUNCH

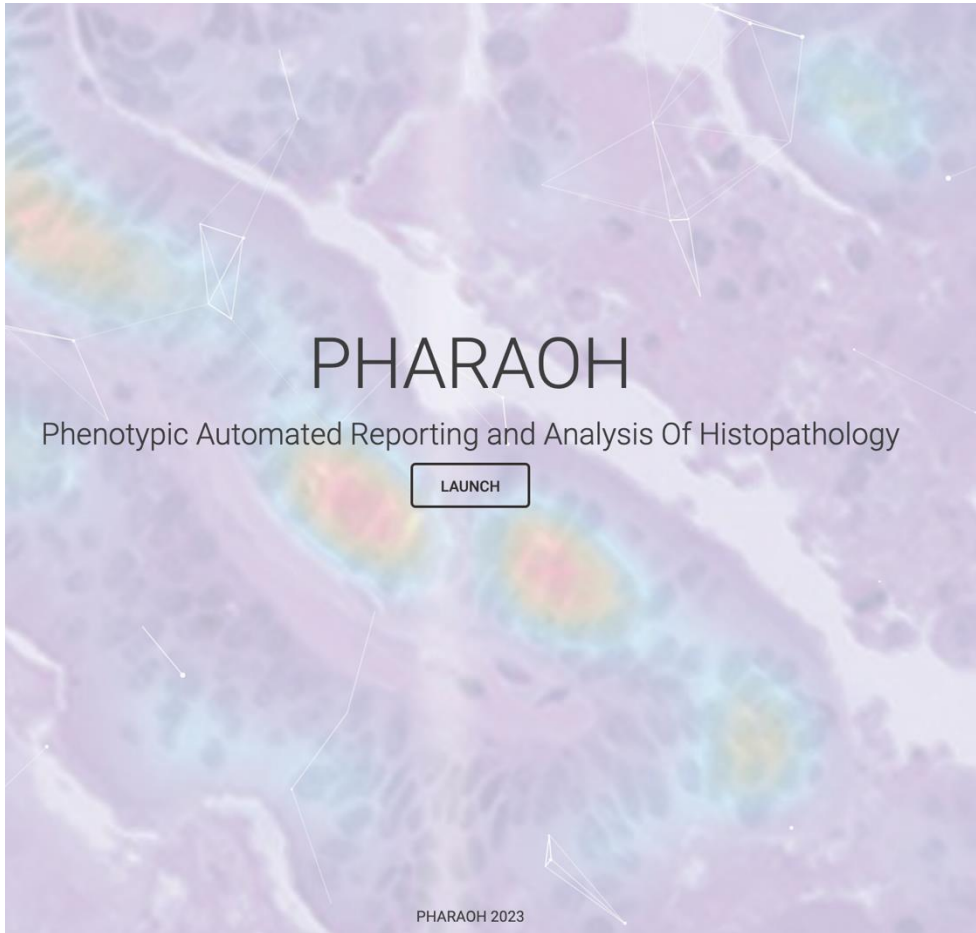
PHARAOH is a virtual crowdsourcing platform that empowers pathologists to design and develop custom AI tools without ever having to code. By sharing creations within the PHARAOH community, we hope to create a comprehensive catalog of computational pathology tools

Currently supported sites 



<https://pathologyreports.ai>

<https://www.pathologyreports.ai/>



PHARAOH: Crowd Sourcing Computational Pathology

Ovarian Carcinoma



Site: Ovarian

Classes:

Blank Space Fatty Tissue Epithelial Tumor Fibrous Tissue Mixture Of Epithelial Tumor And Fibrous Tissue Necrosis Normal Ovarian Parenchyma Tumor edge Hemorrhage

Description:

This H&E ovarian carcinoma tissue classifier was developed using transfer learning and the VGG19 CNN and trained to recognize ovarian serous carcinoma and other surrounding tissue elements. Annotations were carried out on batches of image tiles (dimensions: 256 x 256 um) grouped using image-based clustering (HAVOC) from 8 publicly available TCGA-OV H&E-stained whole slide images. Validation testing was carried out on non-overlapping image tiles from the same cases.

Designed by: Dr. Noor Alsafwani (King Fahd Hospital of University, Saudi Arabia)

SELECT

Gastrointestinal Adenocarcinoma



Site: GI

Classes:

Connective tissue Edge of Tumor Edge Of Tumor And Inflammatory Cells Epithelial Pattern Neoplastic epithelial pattern Inflammatory Cells Mucin Smooth muscle Acute hemorrhage Blank space Necrosis

Description:

This H&E colorectal carcinoma tissue classifier was developed using transfer learning and the VGG19 CNN and trained to recognize colorectal adenocarcinoma and other surrounding tissue elements. Annotations were carried out on batches of image tiles (dimensions: 256 x 256 um) grouped using image-based clustering (HAVOC) from 8 publicly available TCGA-COAD H&E-stained whole slide images. Validation testing was carried out on non-overlapping image tiles from the same cases.

Designed by: Dr. Mohamed Al-Yousef (King Fahd Hospital of University, Saudi Arabia)

SELECT

Common Brain Tumors and Surrounding Tissue Elements



Site: Brain

Classes:

Acute hemorrhage Blank space Cerebellar cortex Chronic hemorrhage Crush Artifact Fibrocollagenous tissue Glial histology Gray matter Normal hematolymphoid tissue Meningothelial Histology Epithelial pattern Skeletal muscle Necrosis Radiation Necrosis Schwannian/spindled Histology White matter

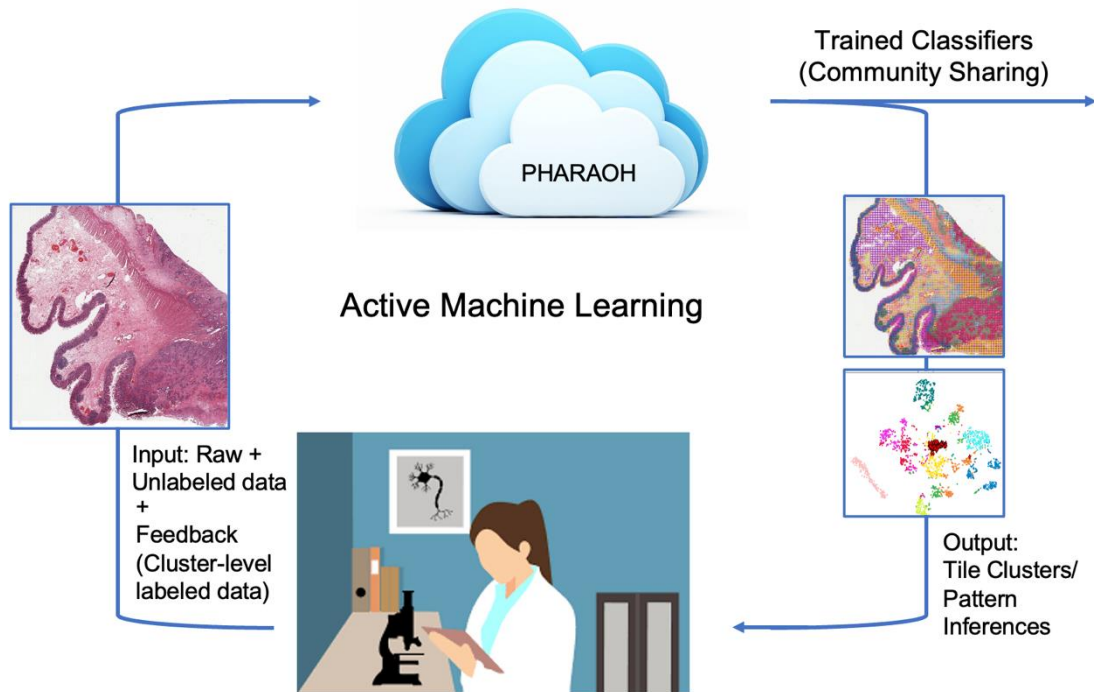
Description:

This H&E brain tumor classifier was developed using transfer learning and the VGG19 CNN and trained to recognize the four most common brain tumor types and other surrounding tissue elements. H&E-stained images patches (dimensions: 512 x 512 um) and annotations were taken from previous publicly available datasets (PMID: 35156037). Validation testing was carried out on non-overlapping image tiles from independent cases.

Designed by: Kevin Faust & Michael Lee and Colleagues (University Health Network, Canada)

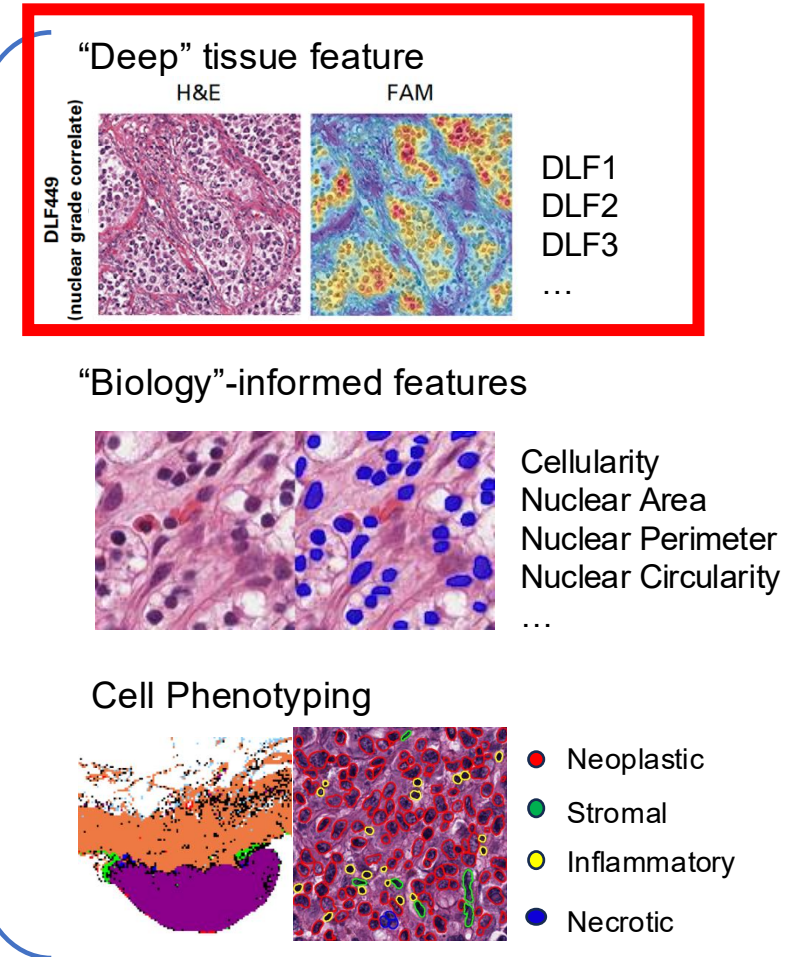
SELECT

Crowdsourcing Classifier Production

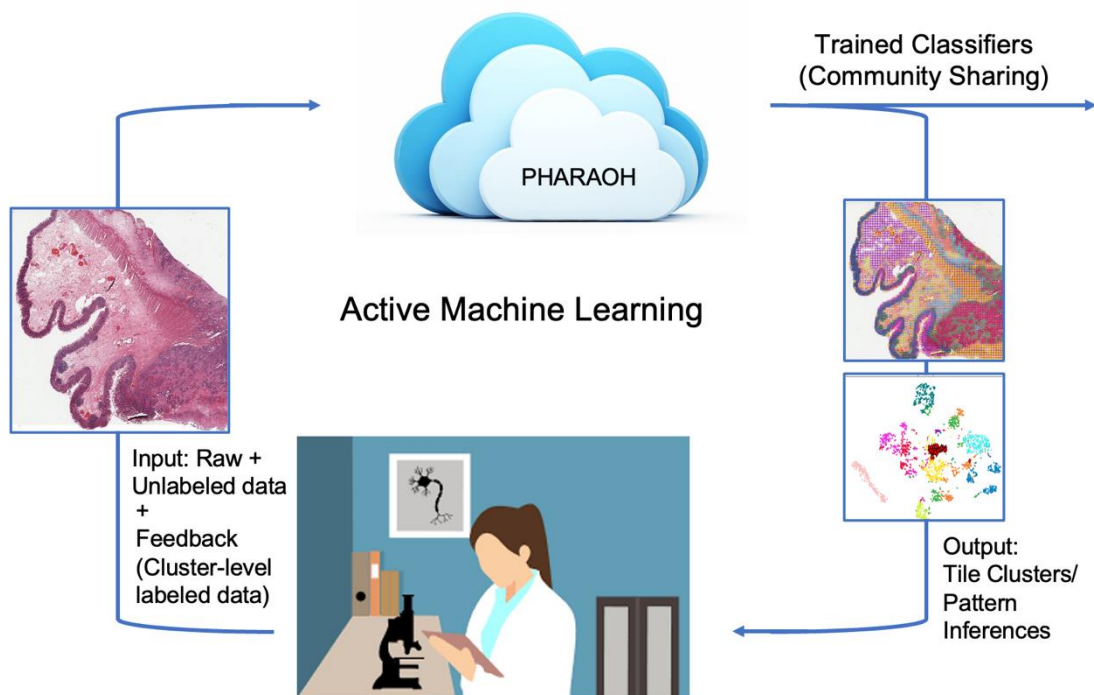


Biomarker (Feature) extraction

Outputs: Phenotyping and Regional Analysis of Histology

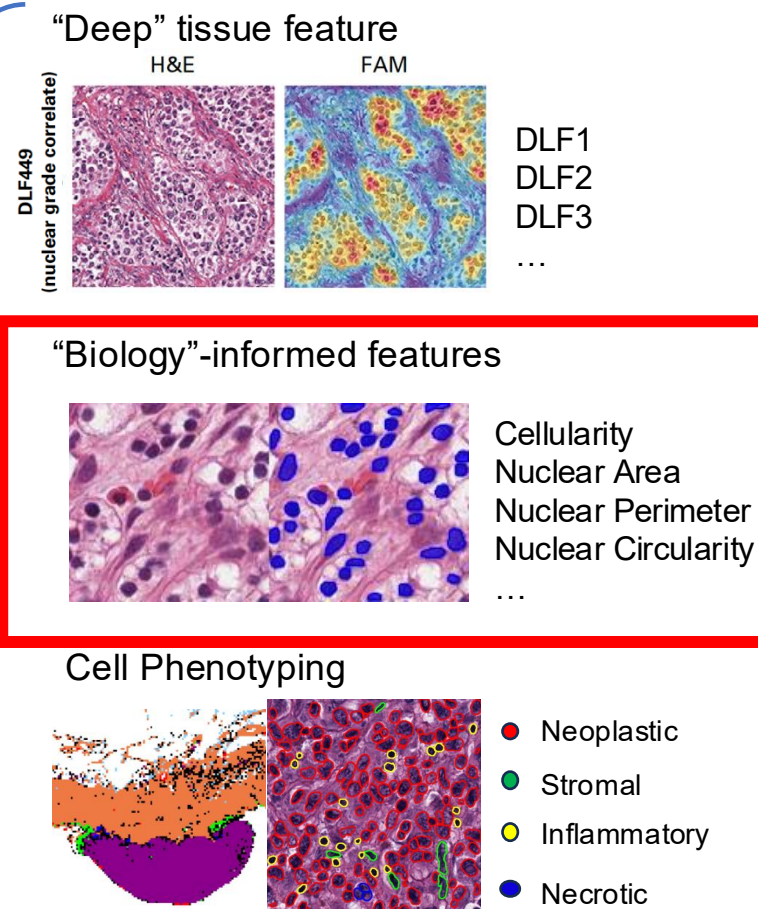


Crowdsourcing Classifier Production



Biomarker (Feature) extraction

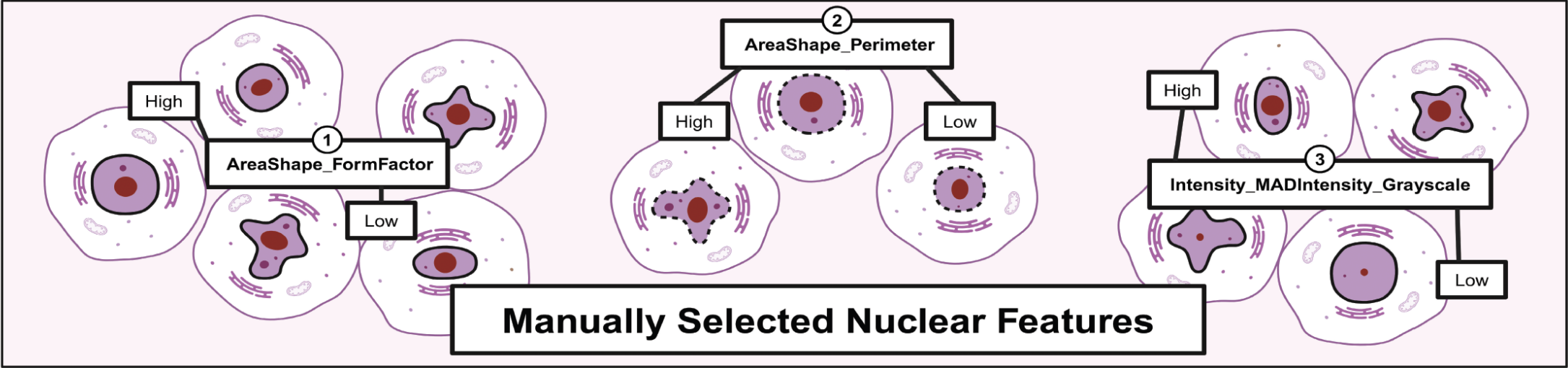
Outputs: Phenotyping and Regional Analysis of Histology



PHARAOH Applications – Tumor Grade automation

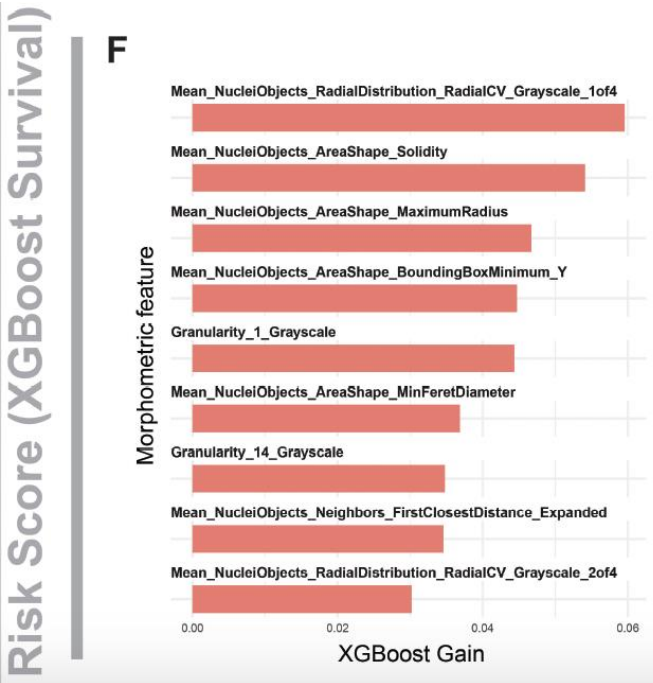
3-Feature Aggregate

A

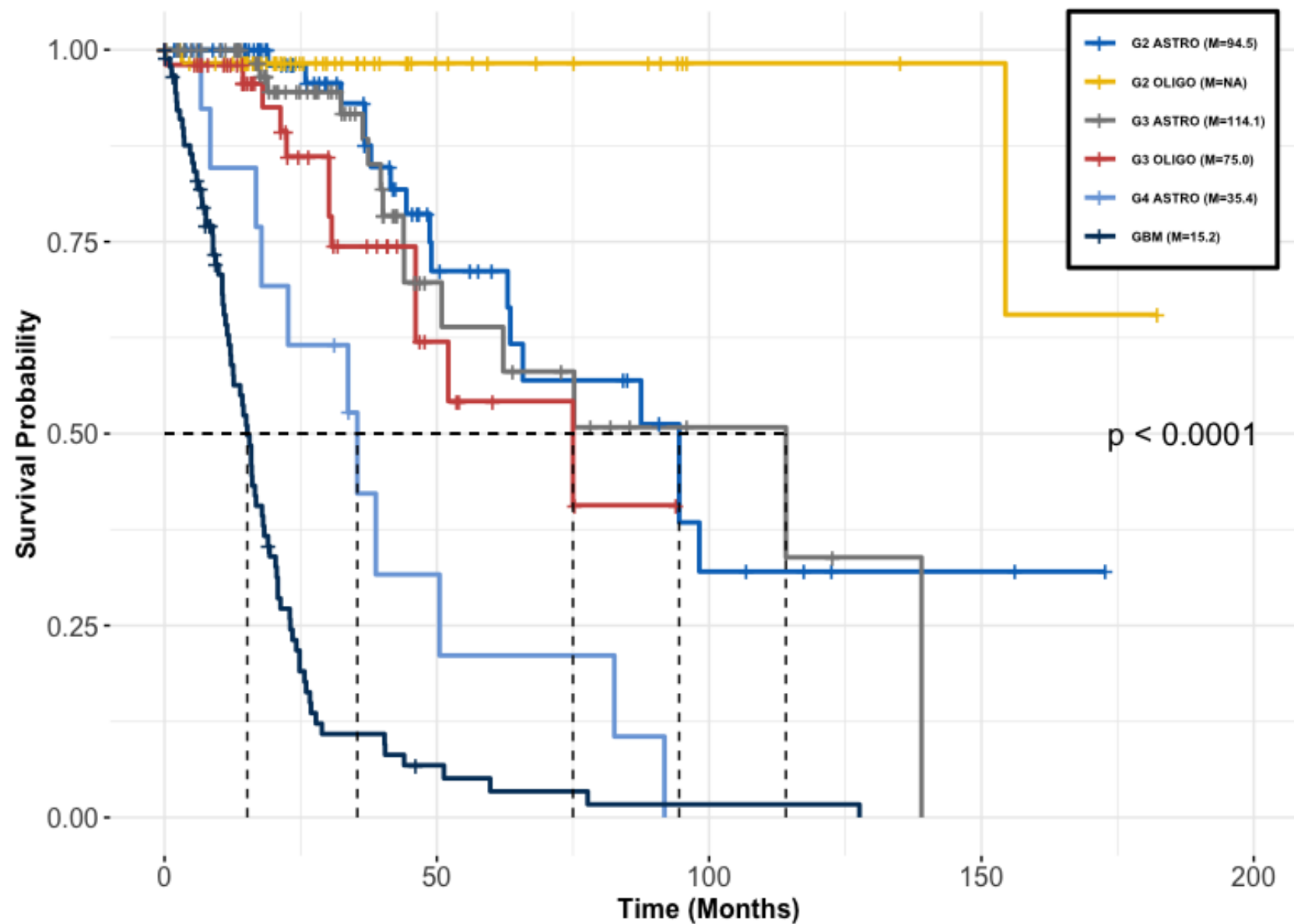


Grade	Nuclei		Nucleoli
	Size (µm)	Shape	
1	10	Round, Uniform	Inconspicuous or absent
2	15	Slightly irregular	Evident at high power (X400 magnification)
3	20	Obviously irregular	Prominent, large at low power (X100 magnification)
4	>20	Bizzare, Often multilobed	Heavy chromatin clump

PHARAOH Applications – Data driven risk score design

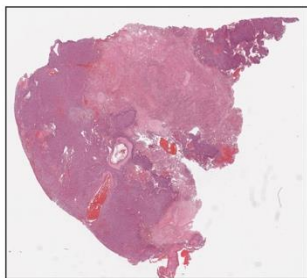


Risk Prediction: IDH-mutated 1p19q code1 Oligodendrogliomas, WHO Gr 3

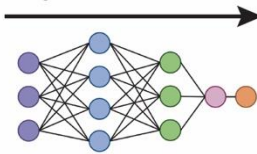


Risk Prediction: IDH-mutated 1p19q code1 Oligodendrogliomas, WHO Gr 3

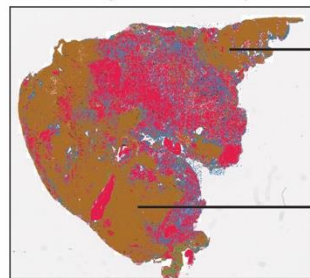
Input for HAVOC-derived AI:
H&E WSI of tumor section



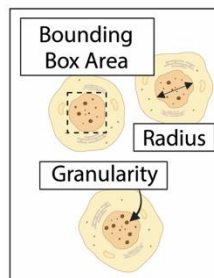
Fine-tuned brain tumor
segmentation CNN model



Output for HAVOC-derived AI:
Tumor segmentation (brown)



CODIDO Outputs
160 Nuclear Features



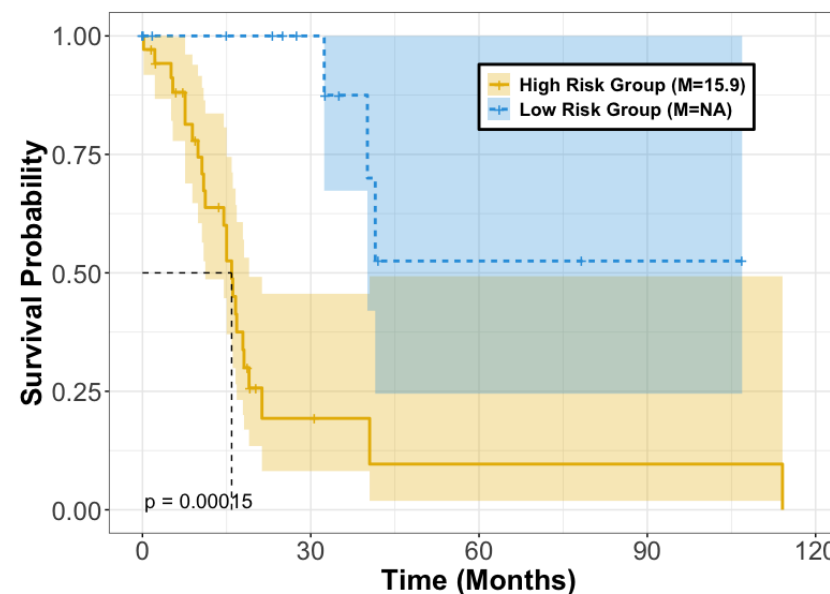
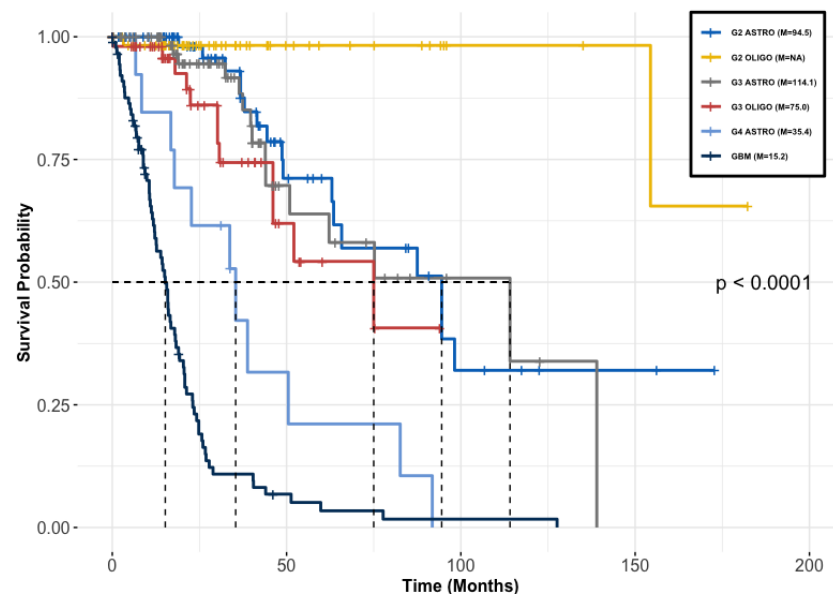
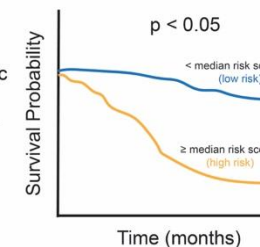
Input:
Nuclear Features
+
Survival Data

XGBoost Survival

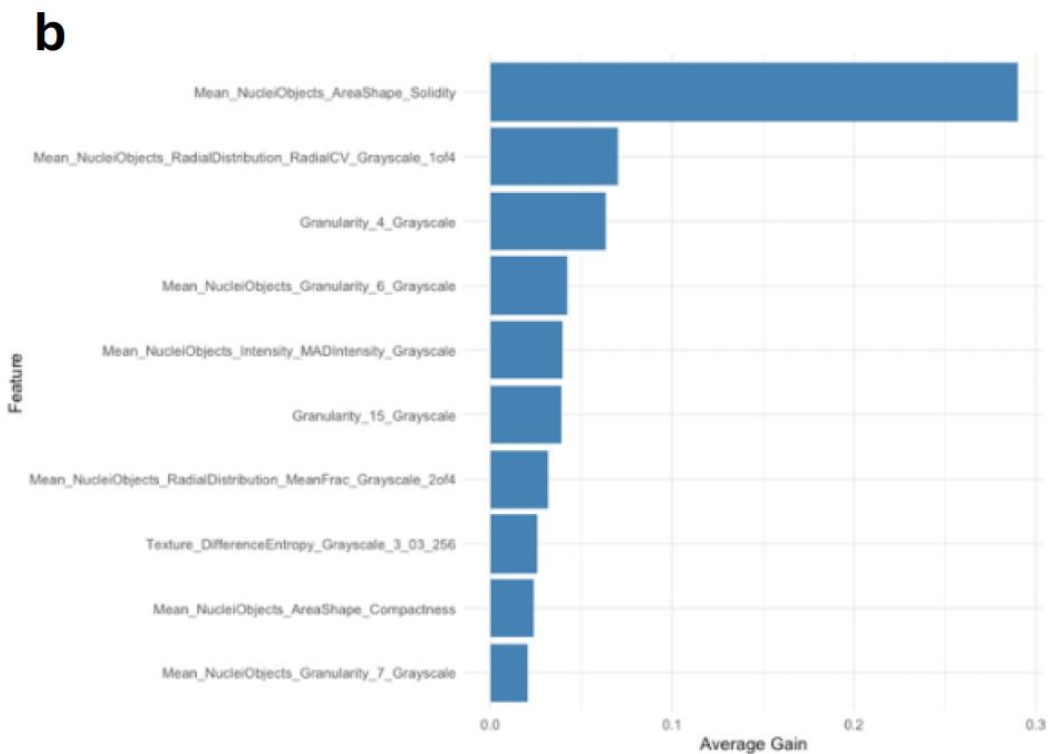
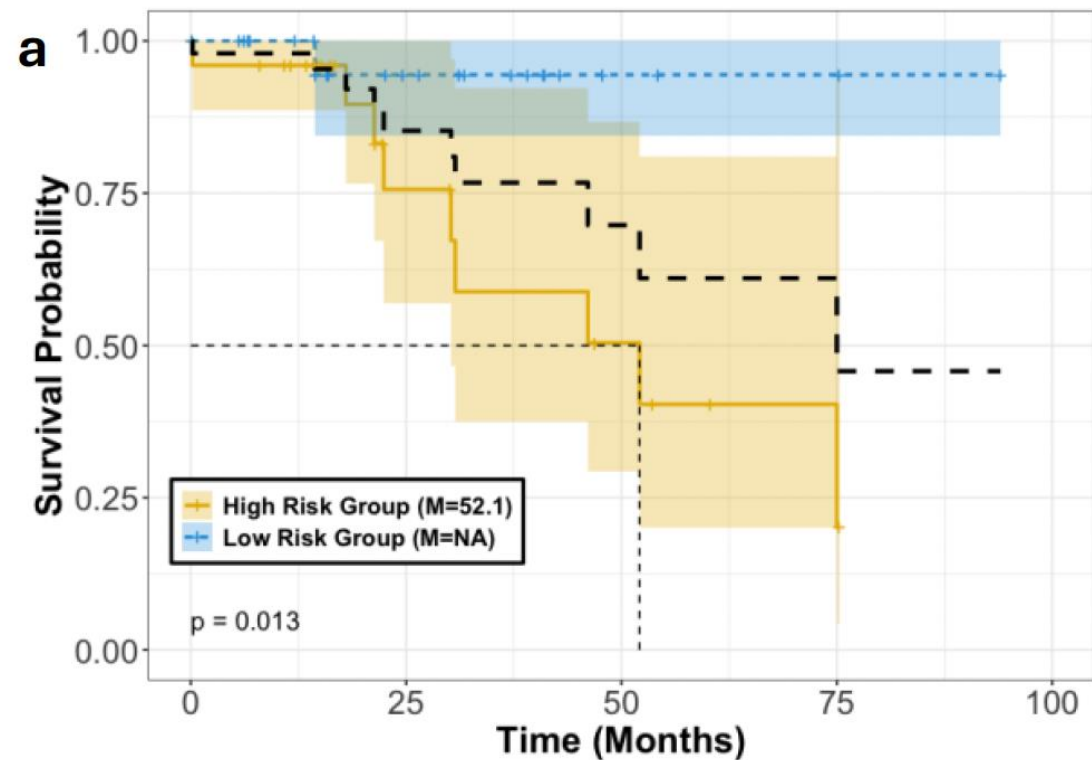


Output:
Multi-parametric
risk signature

Kaplan-Meier
Survival Curve

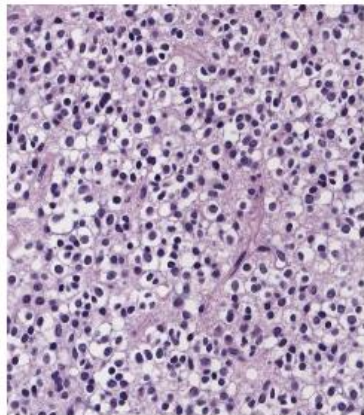


Risk Prediction: IDH-mutated 1p19q code1 Oligodendrogliomas, WHO Gr 3

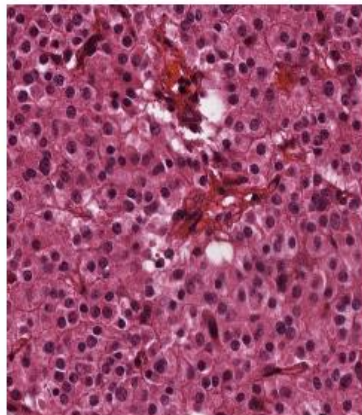


Risk Prediction: IDH-mutated 1p19q code1 Oligodendrogliomas, WHO Gr 3

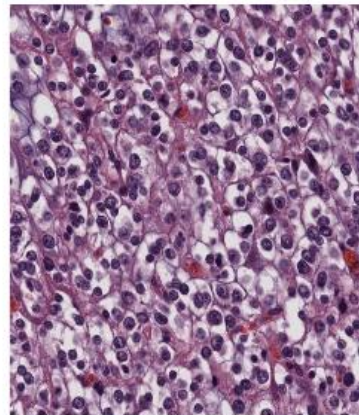
Area Solidity (HIGH)



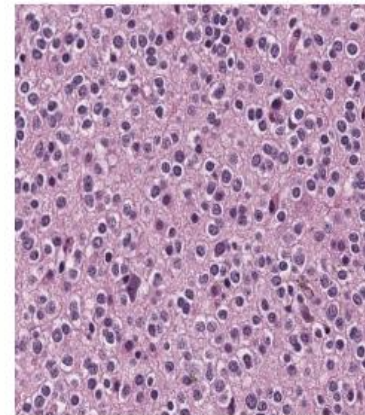
TCGA-DH-5144



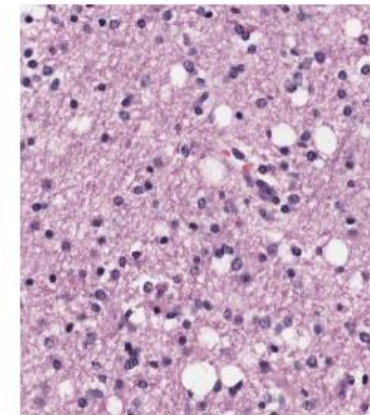
TCGA-S9-A6TW



TCGA-HT-A5R9

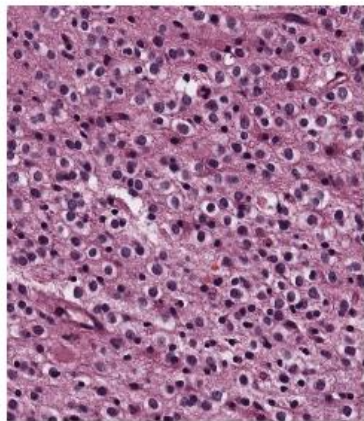


TCGA-DB-A64P

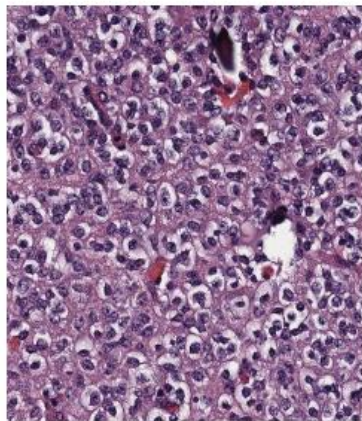


TCGA-FG-7638

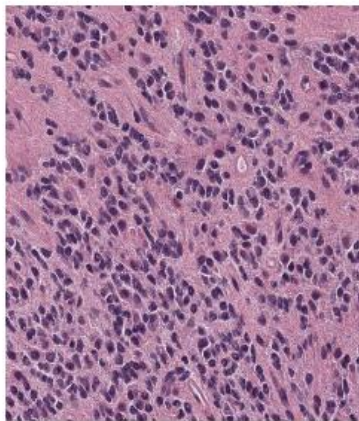
Area Solidity (LOW)



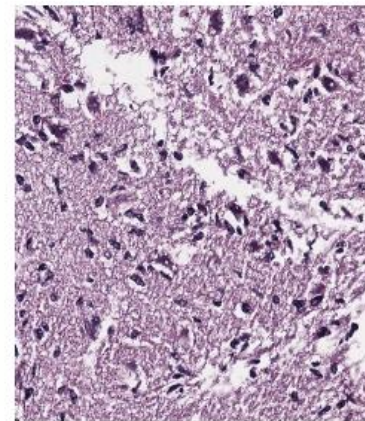
TCGA-HT-7677



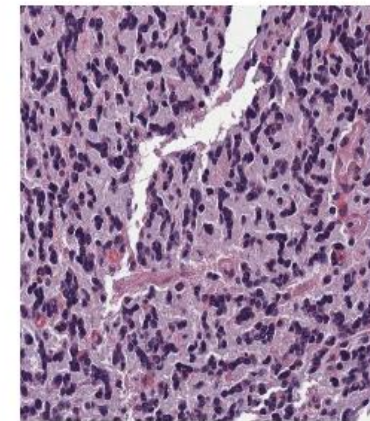
TCGA-DU-6410



TCGA-DU-6394

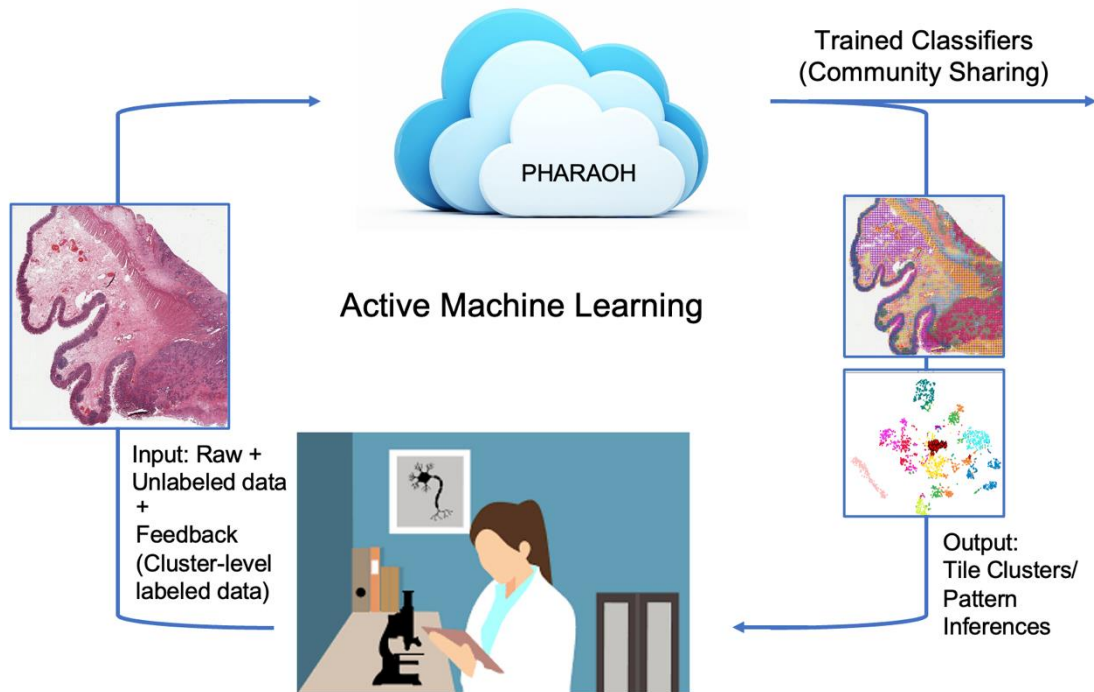


TCGA-DU-7302



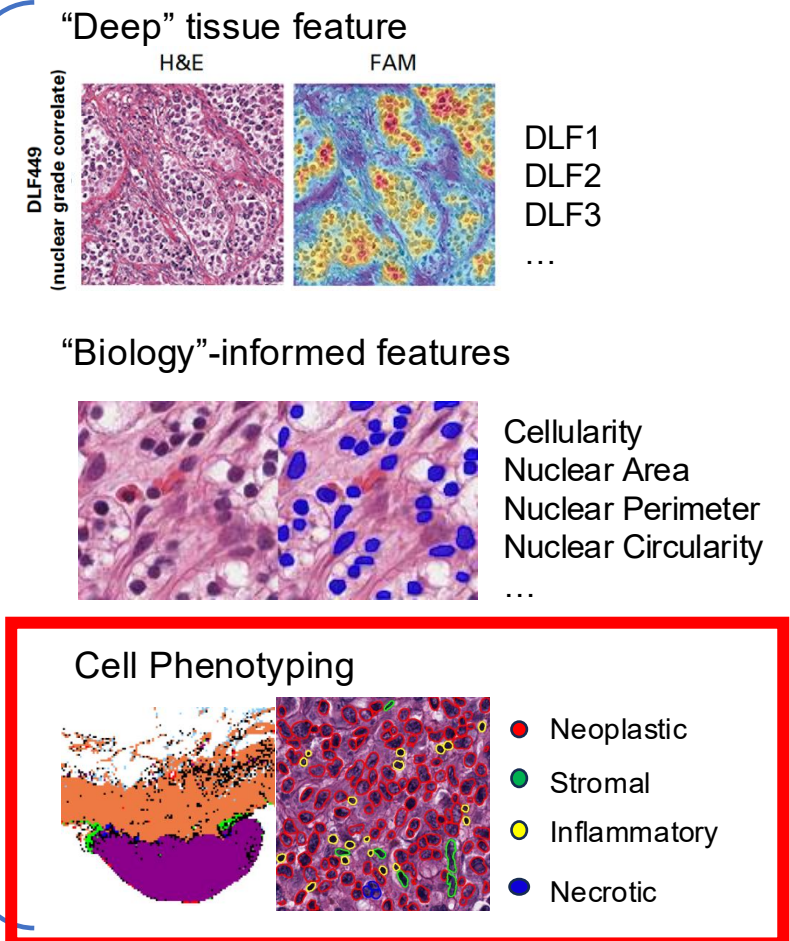
TCGA-DU-6393

Crowdsourcing Classifier Production



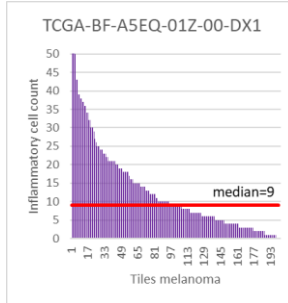
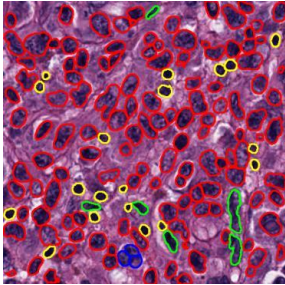
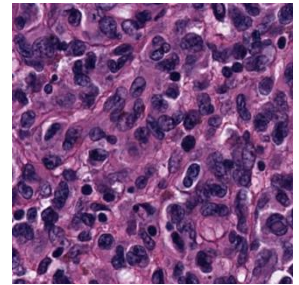
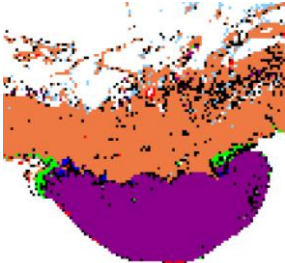
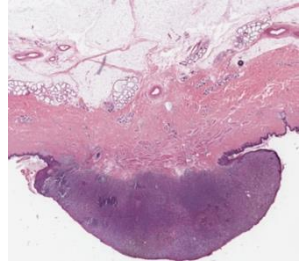
Biomarker (Feature) extraction

Outputs: Phenotyping and Regional Analysis of Histology



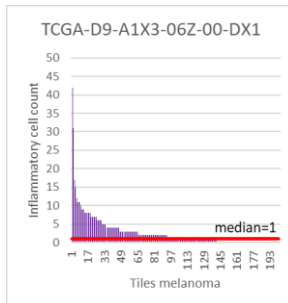
PHARAOH: Quantitative prognostic biomarker design

TCGA-BF-A5EQ-01Z-00-DX1

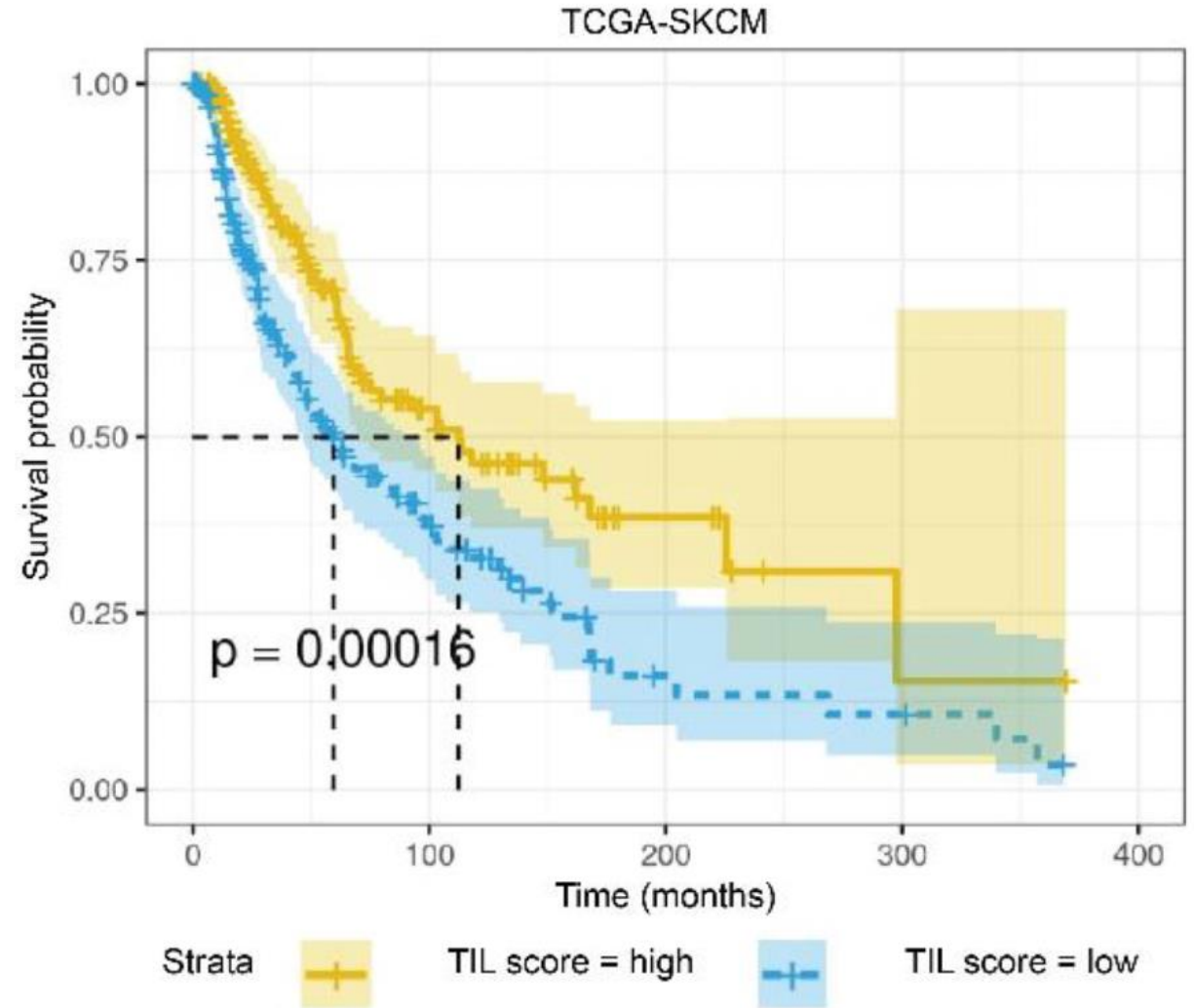
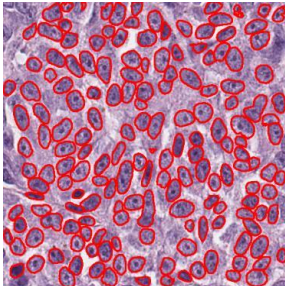
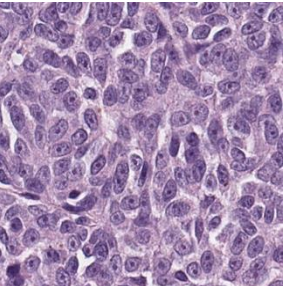
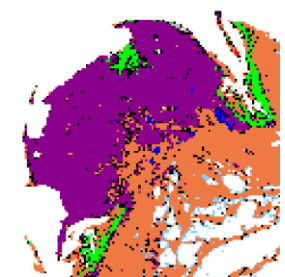
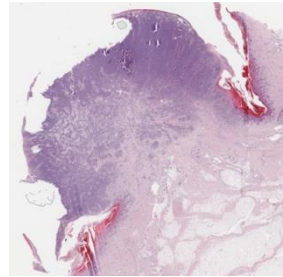


class

- adipose
- connective_fibrosis
- epithelium_melanocyte
- lymph_dense
- melanoma
- necrosis
- unknown
- blood

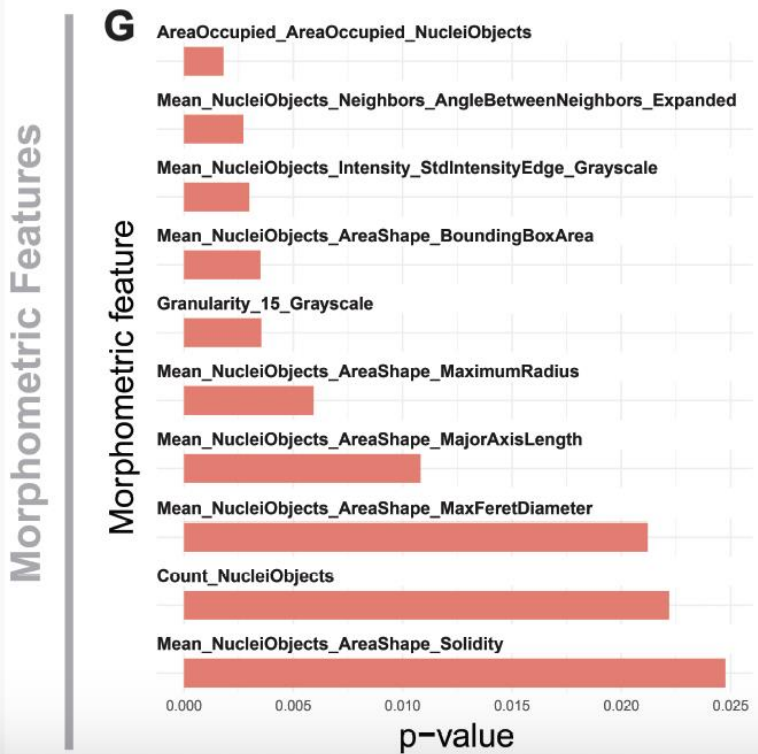


TCGA-D9-A1X3-06Z-00-DX1

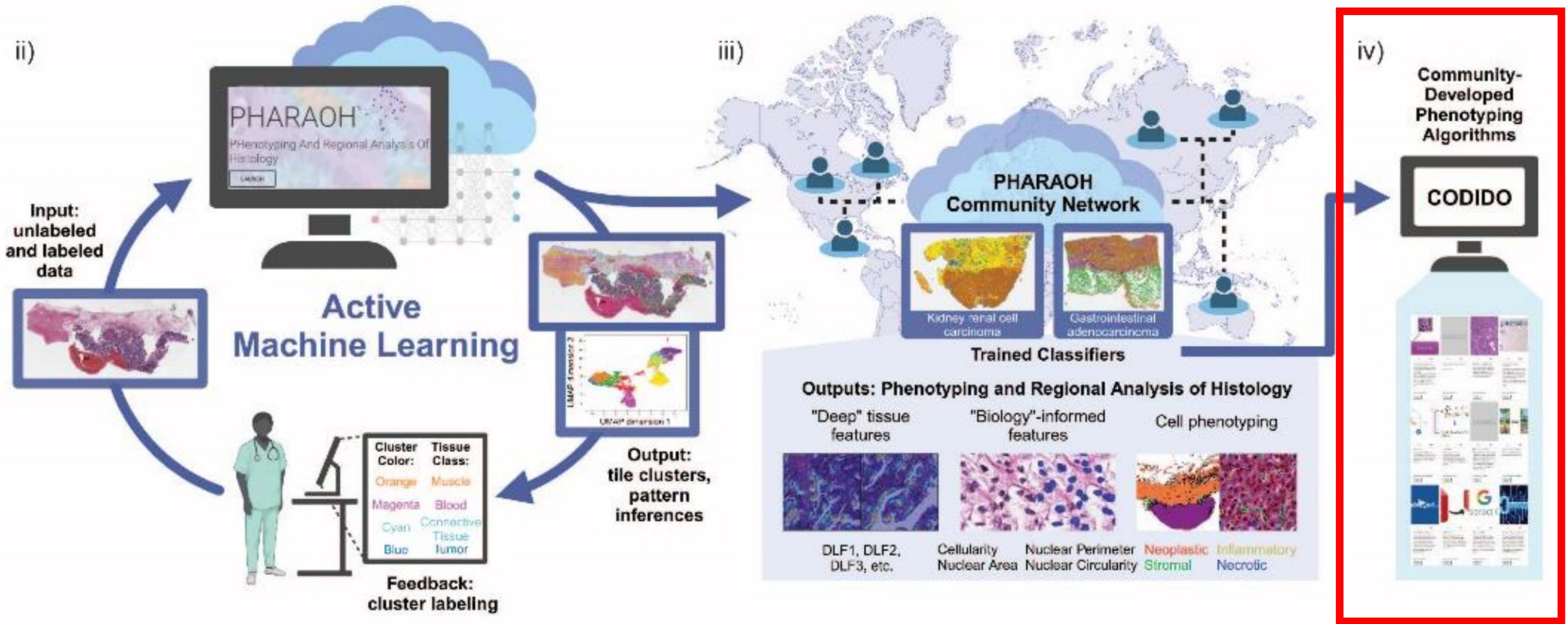


PHARAOH: Biological exploration and discovery

Mitotic Spindle Activity

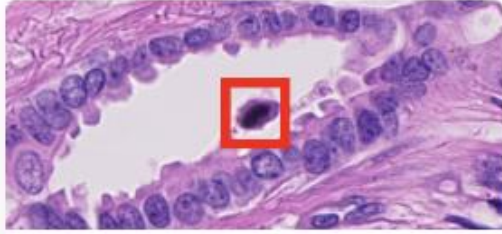


PHARAOH: Crowdsourcing Clinical Expertise



Scaling Computational Pathology through crowdsourcing (codido.co)

Mitosis Detection



0

60

mitotic figure count

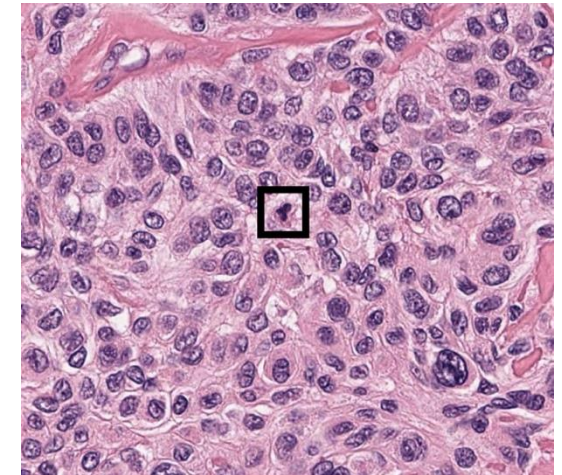
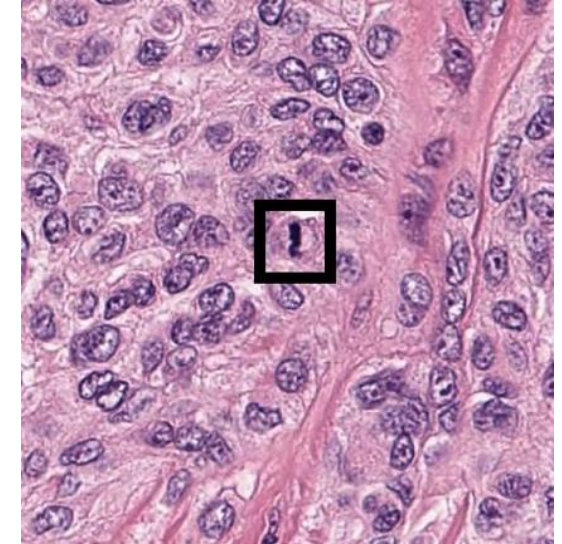
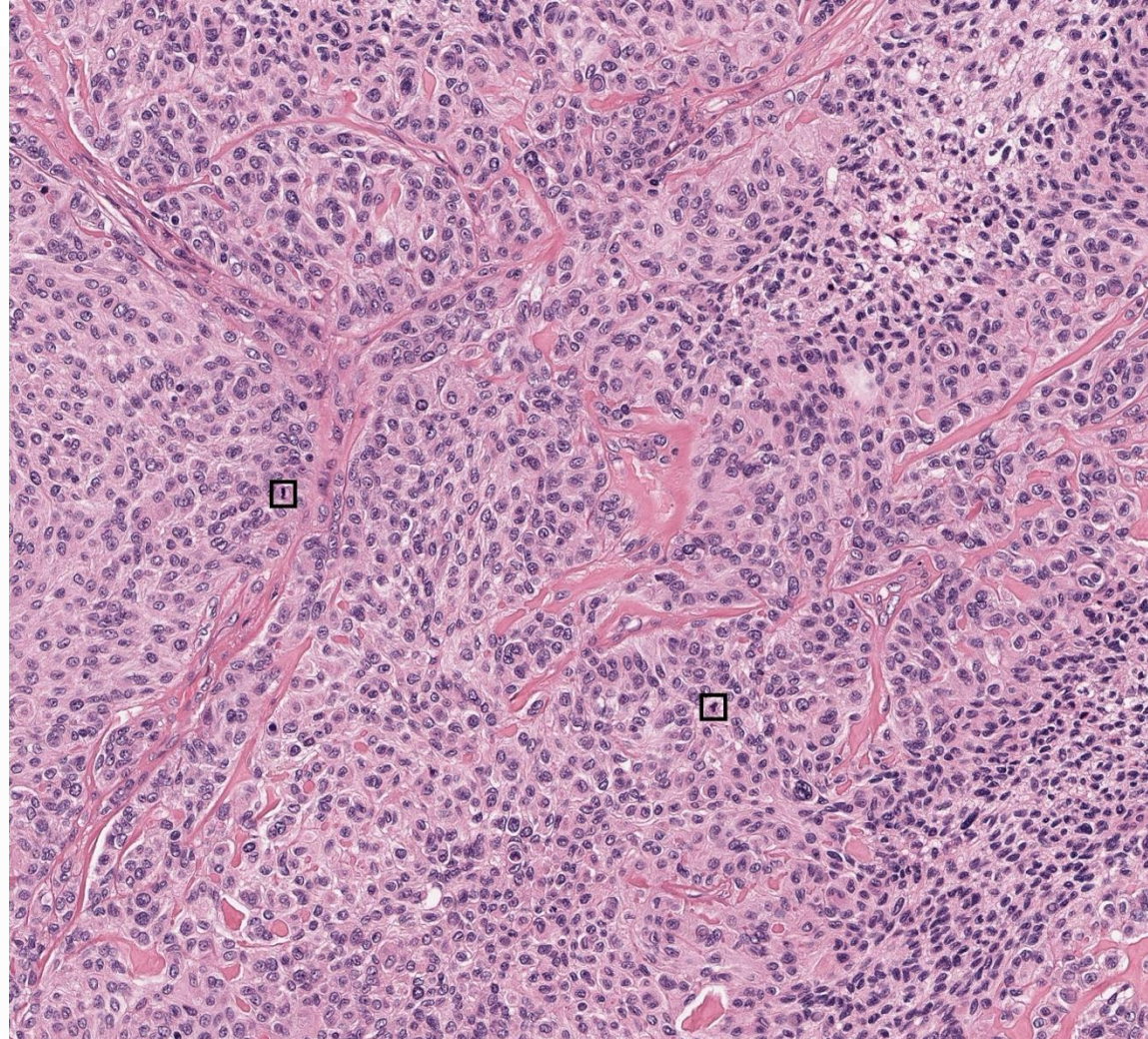
input image or zip with multiple
image files. mpp = microns per pixel
mm2pfov = mm2 per feild of view
based on:

<https://www.nature.com/articles/s41597-023-02327-4>

<https://github.com/DeepMicroscopy/MIDOGpp/tree/main?tab=readme-ov-file>

Published by Andrew Young

SELECT



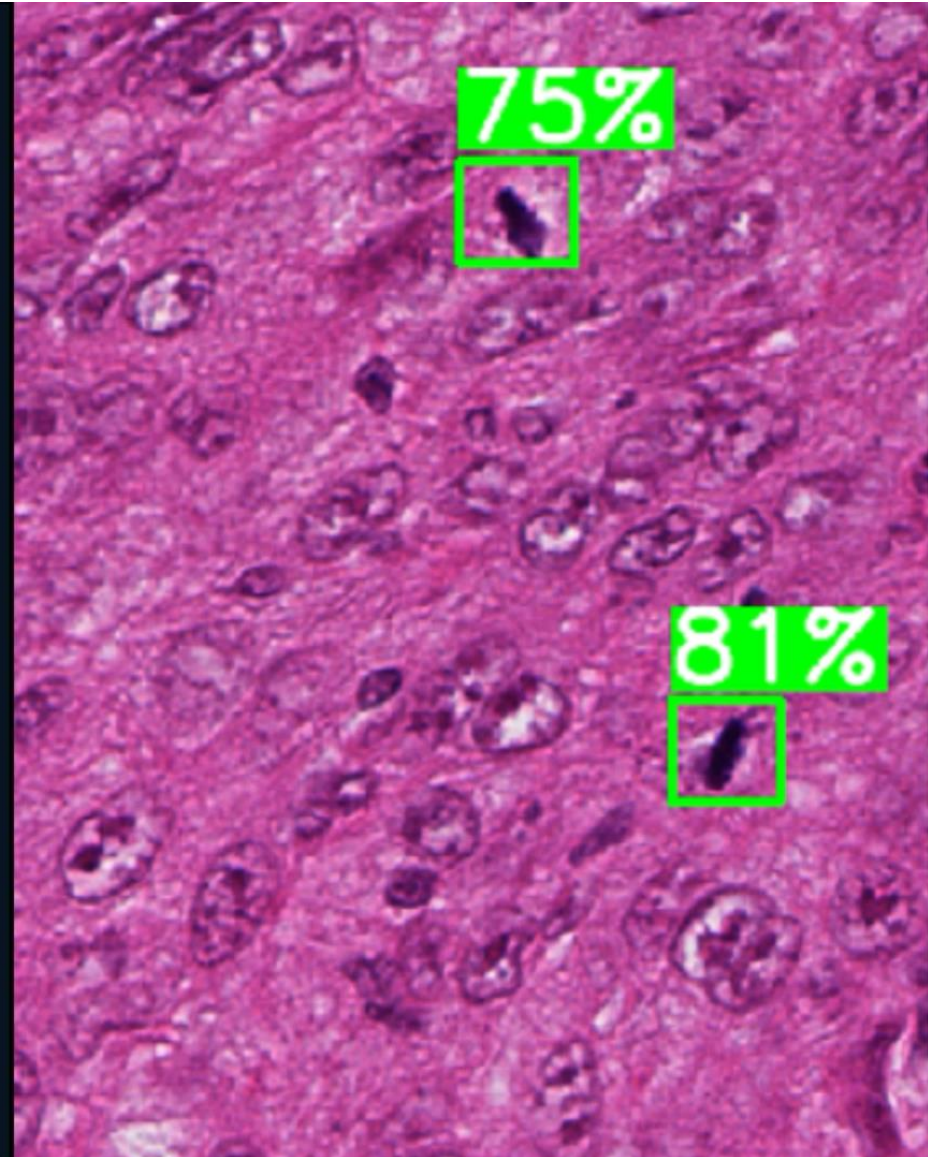
OnSight Pathology: Software for real world deployment

Onsight: A real-time computational pathology companion for histopathology

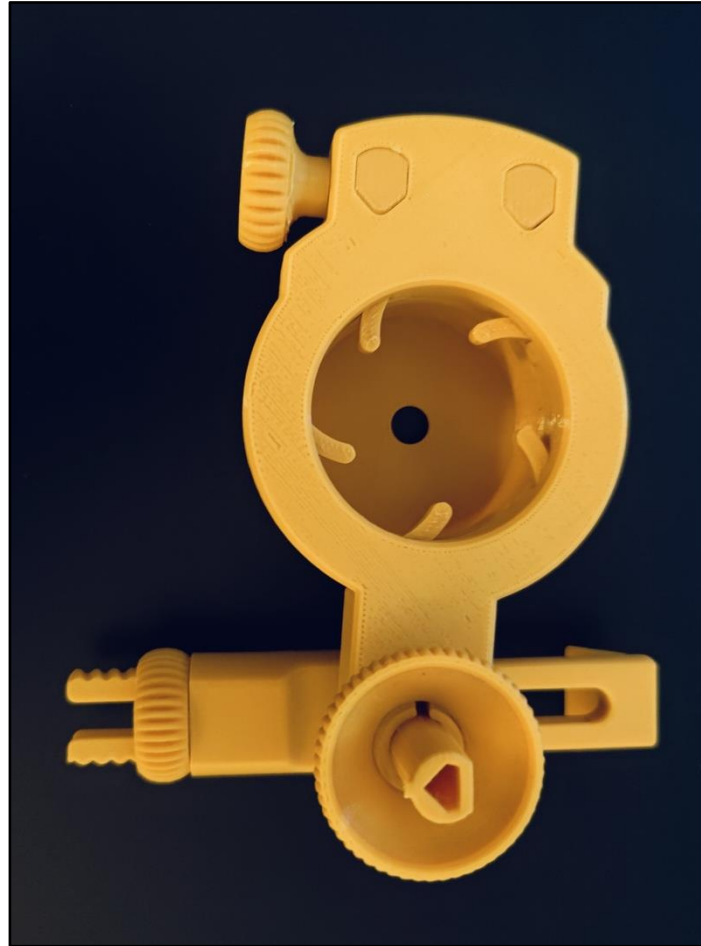
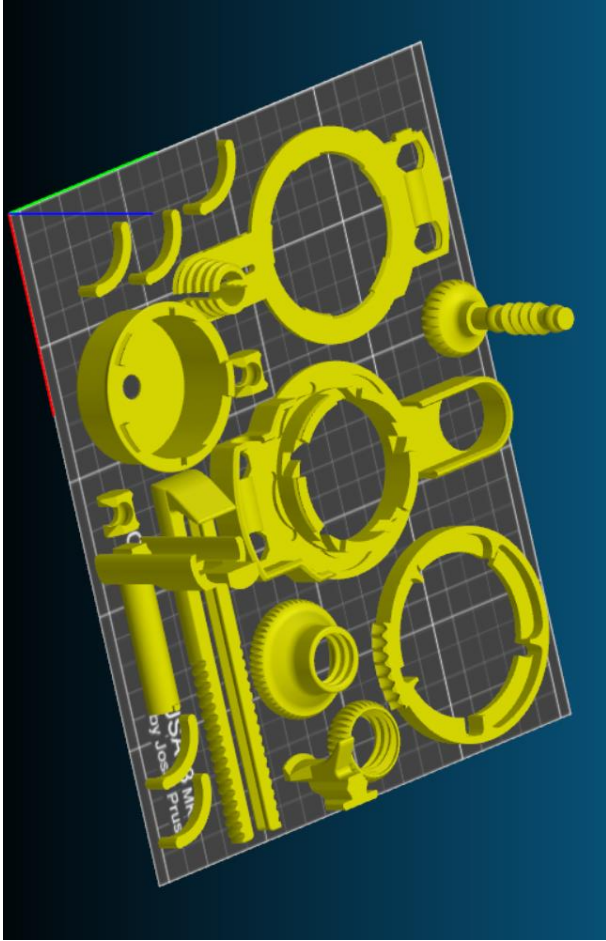


University of Toronto

Princess Margaret Cancer Center

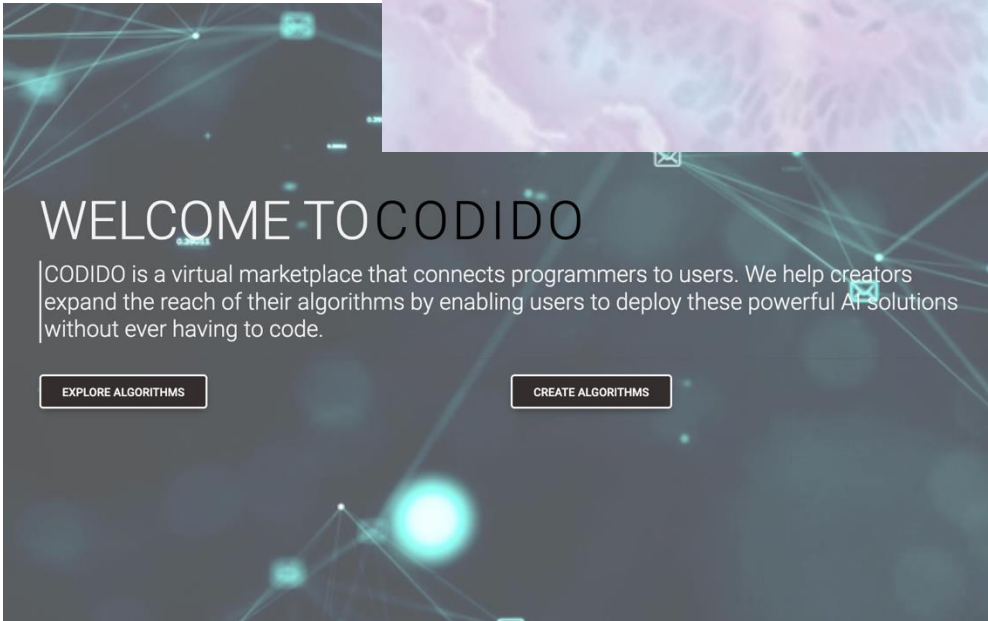
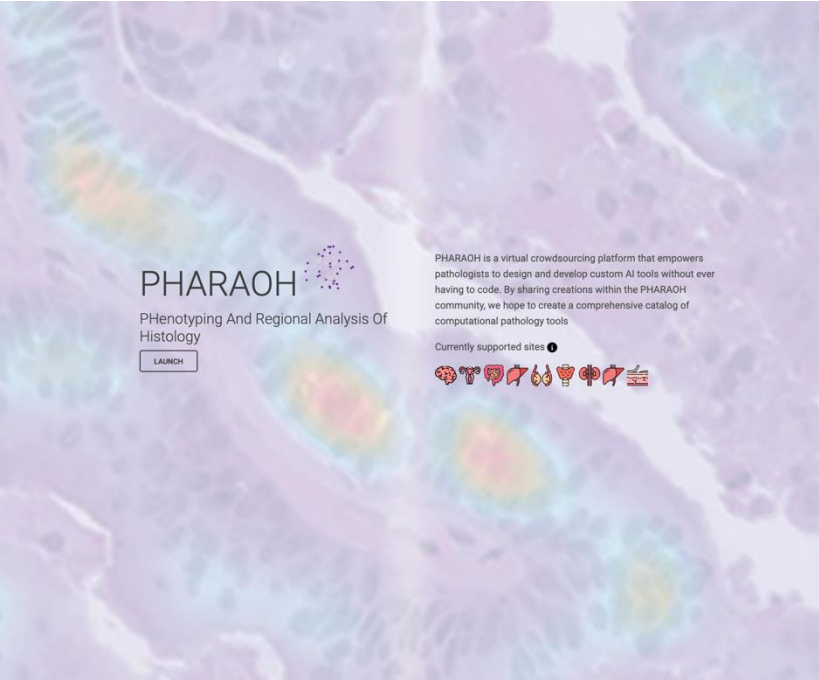


OnSight Pathology: Software for real world deployment



Credit: Shane Eaton PGY-4, NP

Safety Considerations / Peer-Review



CODIDO Marketplace

Here you can find a vast range of algorithms to use.

[\(Looking for an algorithm that we don't have yet? Send your suggestions here!\)](#)

Browse Algorithms

Random FILTER



49

★★★★★

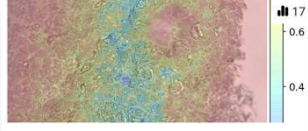
5.0 out of 5 (3 reviews)

Histomic Atlases of Variation of Cancer (HAVOC)

HAVOC is a versatile tool that maps histomic heterogeneity across H&E-stained digital slide images to help guide regional deployment of molecular resources to the most relevant/biodiverse tumor niches. Specify a single or multiple (separated by...

Published by Kevin Faust

SELECT



17

★★★★★

5.0 out of 5 (1 reviews)

Efficient Focus Quality Assessment for Digital Pathology (FocusLiteNN)

Description: A High Efficiency Focus Quality Assessment for Digital Pathology (MICCAI 2020). FocusLiteNN is 100x faster than ResNet50 while maintaining impressive performance. Surprisingly, it only has 148 parameters. Parameters: num_channel...

Published by Zhongling Wang

SELECT



34

★★★★★

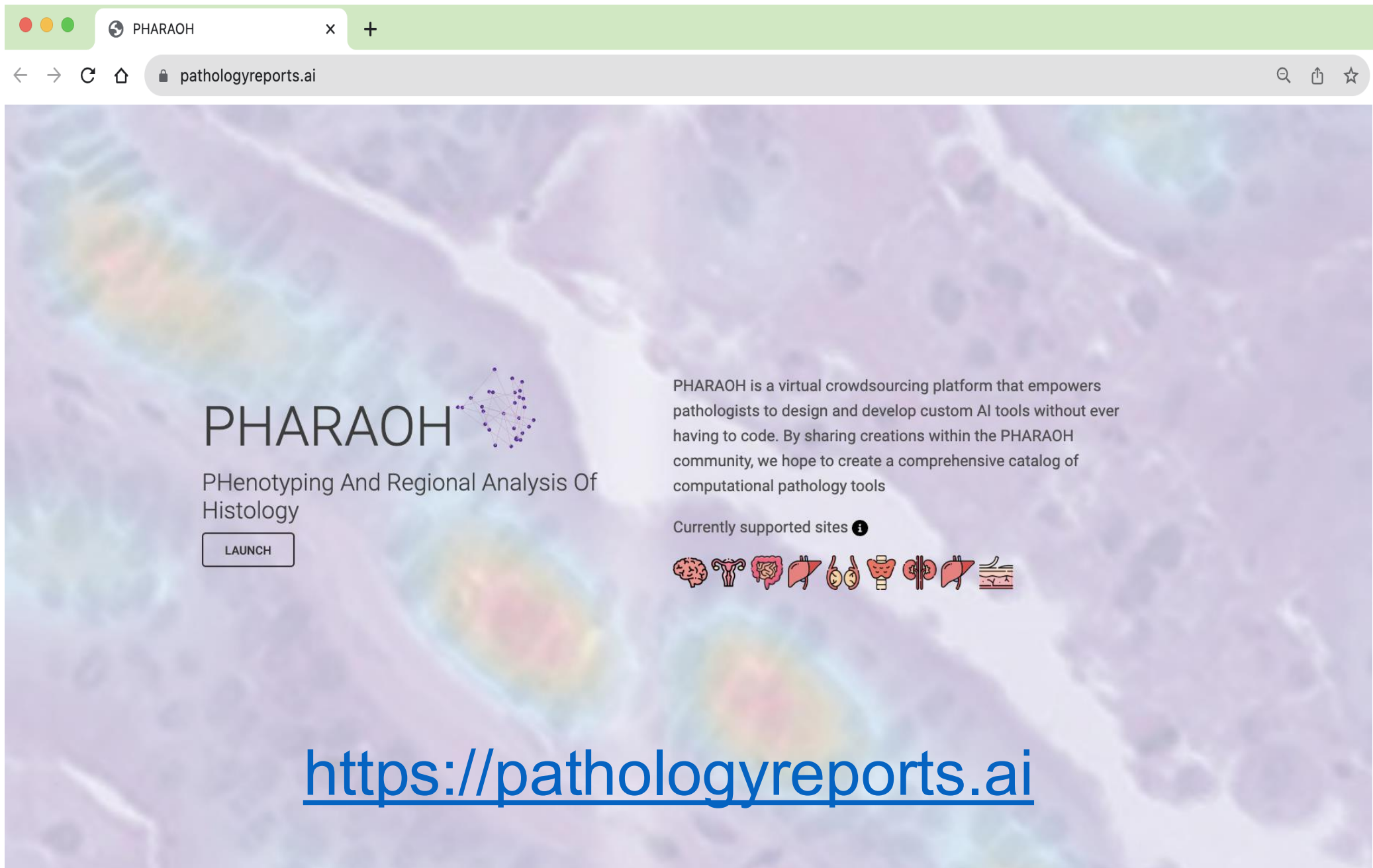
5.0 out of 5 (1 reviews)

Physics-Guided Deep Image Prior network for Stain deconvolution (PGDIPS)

Upload an histopathology image and perform stain deconvolution. Hyperparameters: iters: number of iterations n_stains: number of stains in the uploaded image exc: weight of the exclusion loss, increase this if concentration maps overlap and...

Published by Jianan Chen – Sunnybrook Research Institute

SELECT



PHARAOH



PHeNOTyping And Regional Analysis Of
Histology

LAUNCH

PHARAOH is a virtual crowdsourcing platform that empowers pathologists to design and develop custom AI tools without ever having to code. By sharing creations within the PHARAOH community, we hope to create a comprehensive catalog of computational pathology tools

Currently supported sites 



<https://pathologyreports.ai>

Additional Parameters / Limitations and Caveats

- Adoption has been slow compared to molecular innovations
- Caveats:
 - Staining variation
 - Magnification level
 - Context specific / Concept drift
- New Technologies/Innovations:
 - Vision Transformer (ViT) models
 - Multimodal models
 - Image Search
- Competing technologies:
 - Molecular
 - Spatial



DIAMANDIS LAB

PHARAOH/CODIDO TEAM

Kevin Faust
Minli Chen
Parsa Babaei Zadeh
Alberto Leon
Dimitri Oreopoulos
Bruce Lui
Peter Daun
Paul Daun
Evelyn Kamski-Hennekam

Diamandis Lab

Rifat Sajid
Okty Abbasi Borhani
Laura Sasiadek
Lennart van Winden
Jassy Singh
Rick Sudgen
Ingrid Campbell
Ameesha Paliwal



CIHR IRSC

The Princess Margaret
Cancer Foundation  UHN



Laboratory Medicine & Pathobiology
UNIVERSITY OF TORONTO

Acknowledgements

